

BFS Traversal

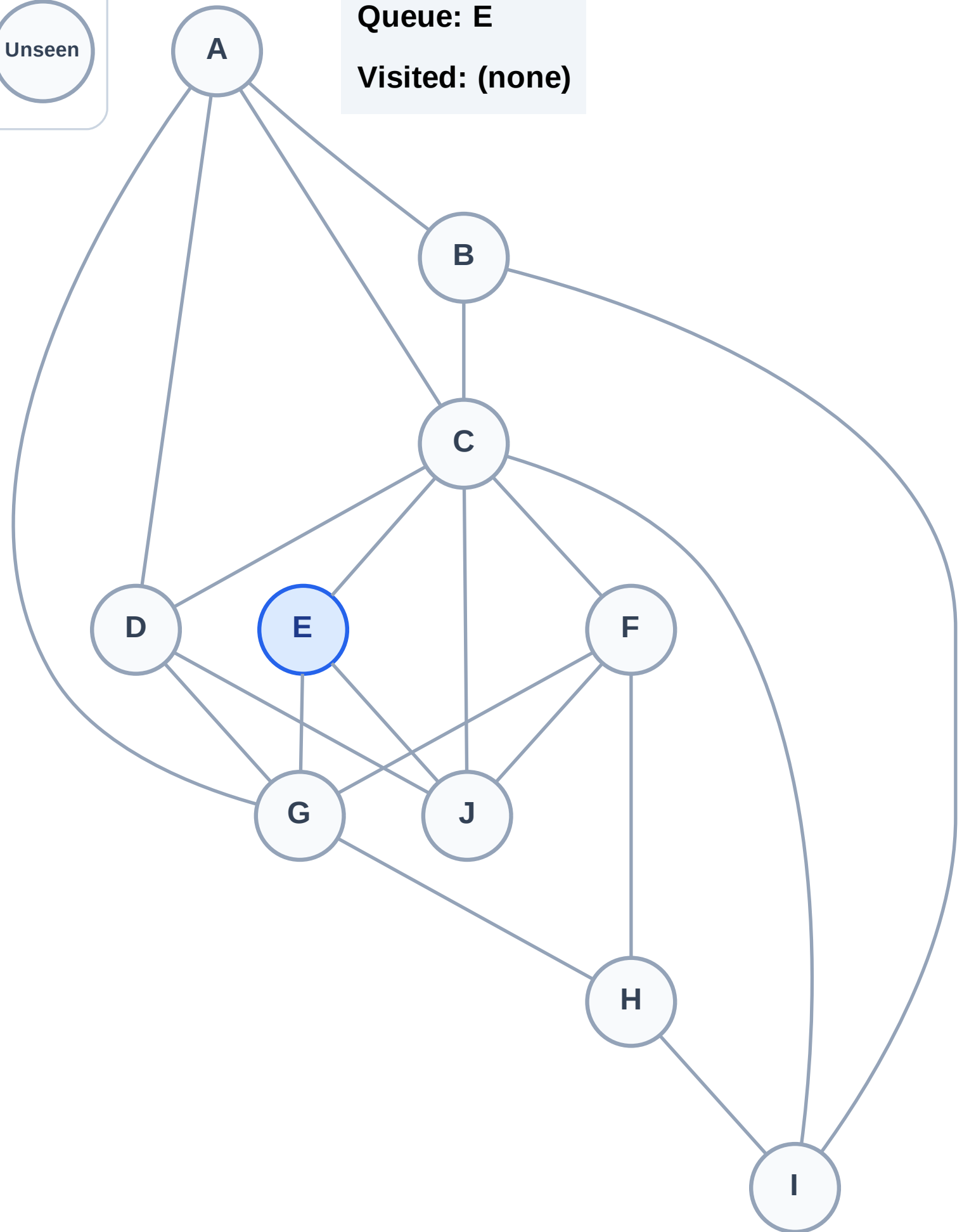
Step 1: Start at E

Legend



Queue: E

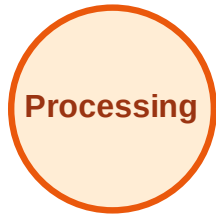
Visited: (none)



BFS Traversal

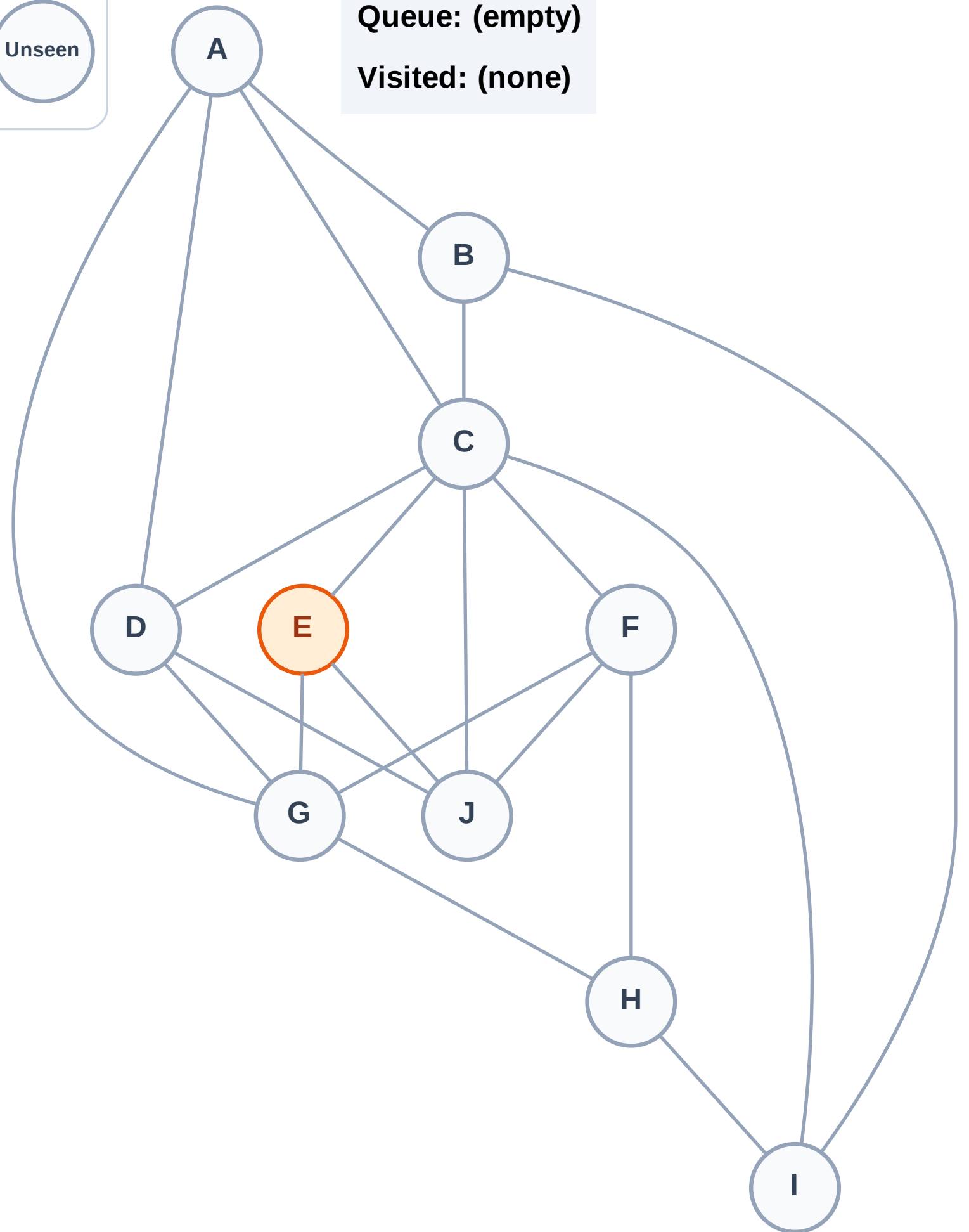
Step 2: Process node E

Legend



Queue: (empty)

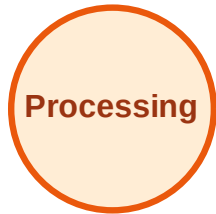
Visited: (none)



BFS Traversal

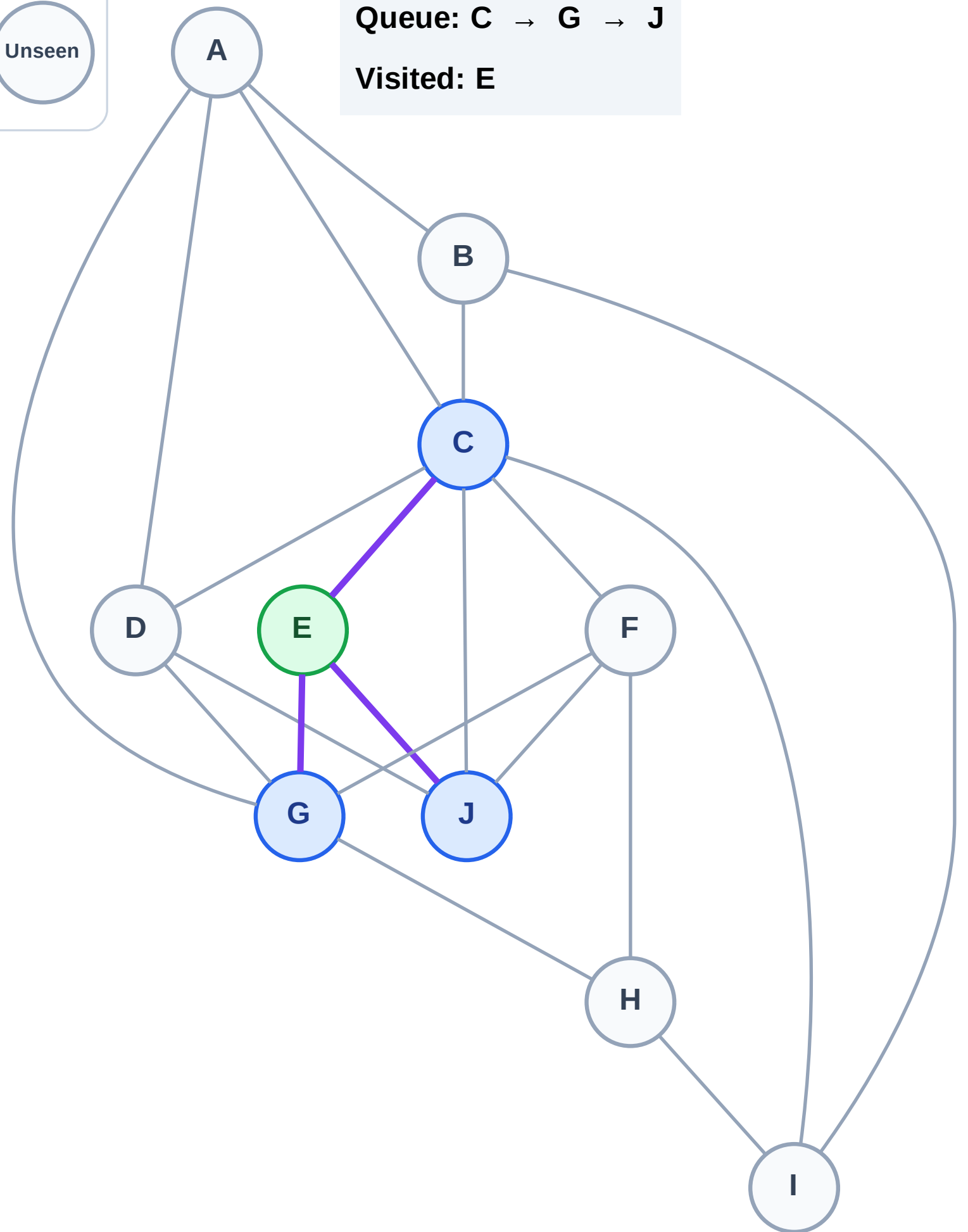
Step 3: Discover C, G, J from E

Legend



Queue: C → G → J

Visited: E



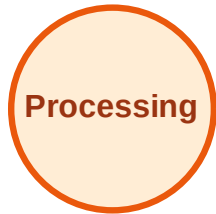
BFS Traversal

Step 4: Process node C

Legend



Complete



Processing



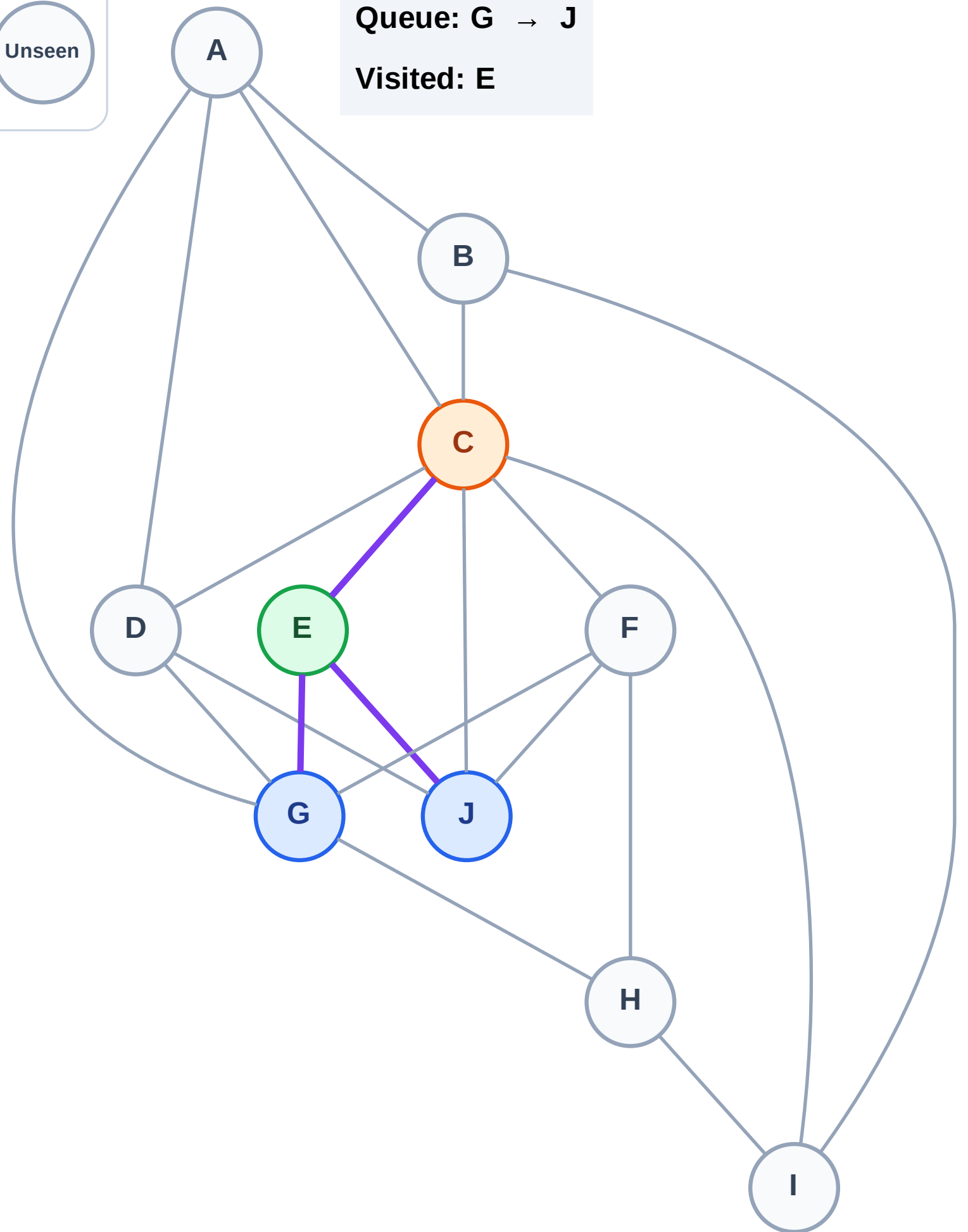
Discovered



Unseen

Queue: G → J

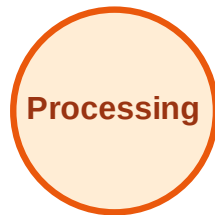
Visited: E



BFS Traversal

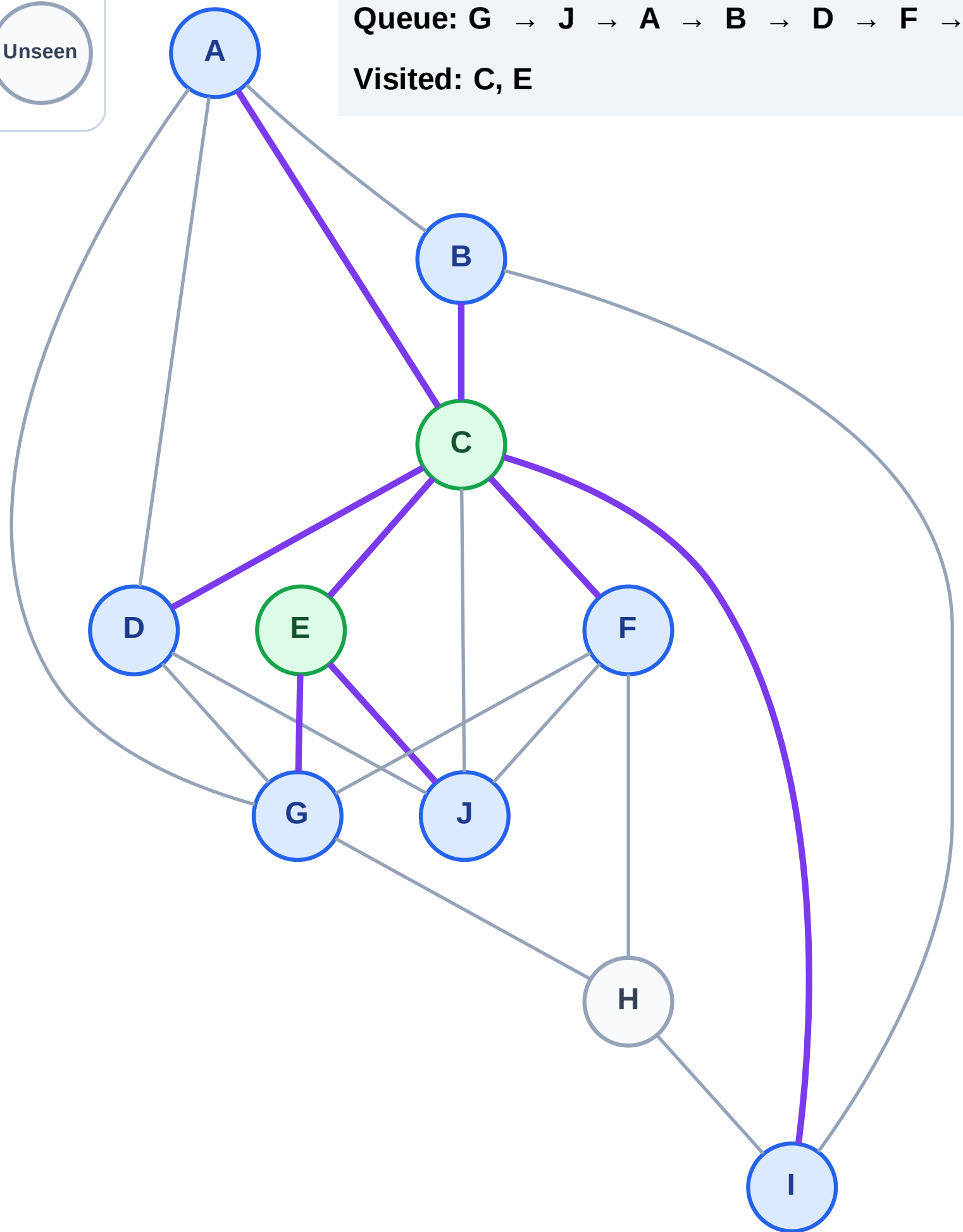
Step 5: Discover A, B, D, F, I from C

Legend



Queue: G → J → A → B → D → F → I

Visited: C, E



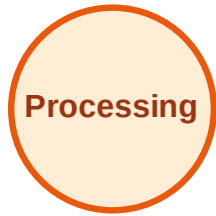
BFS Traversal

Step 6: Process node G

Legend



Complete



Processing



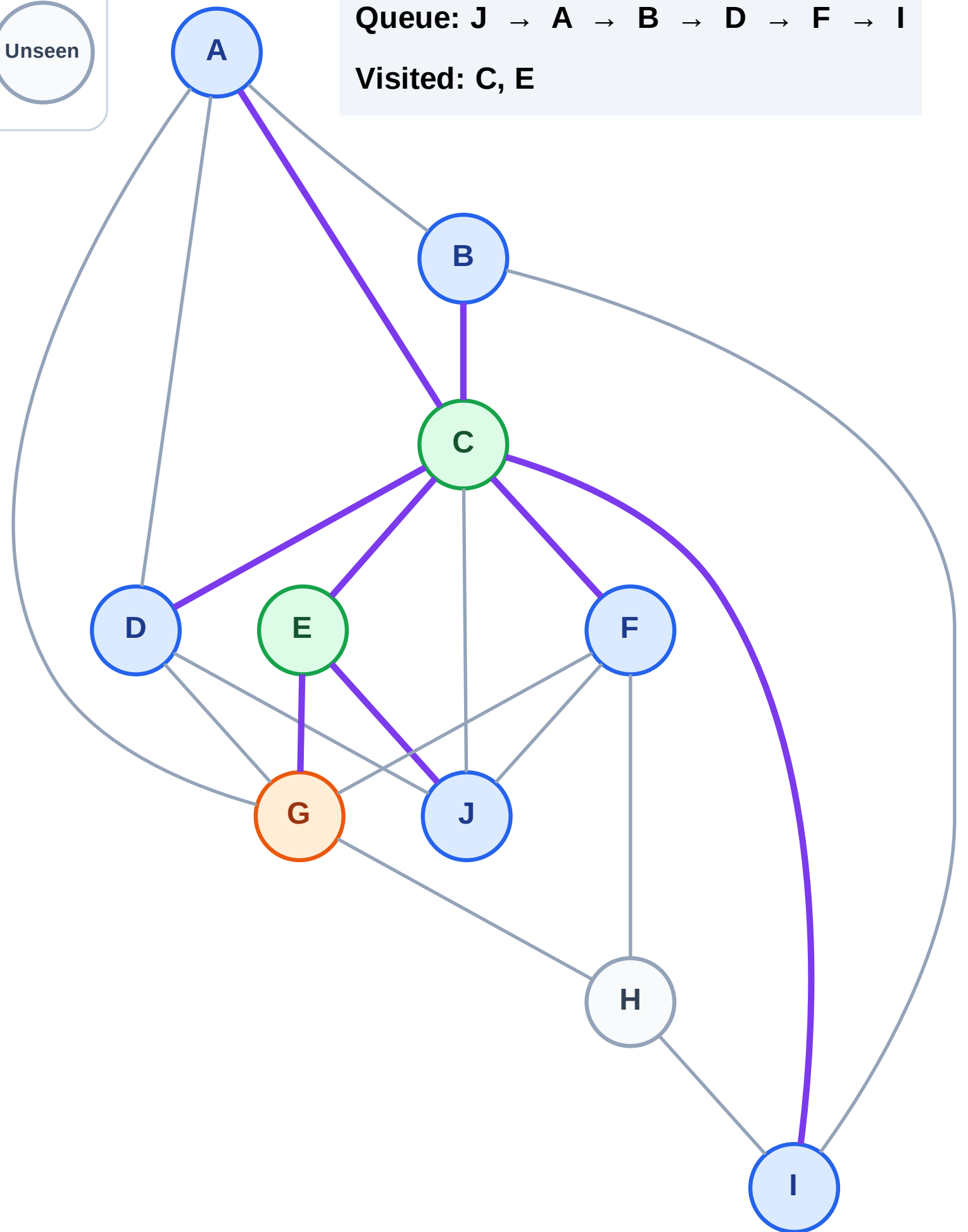
Discovered



Unseen

Queue: J → A → B → D → F → I

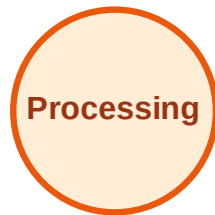
Visited: C, E



BFS Traversal

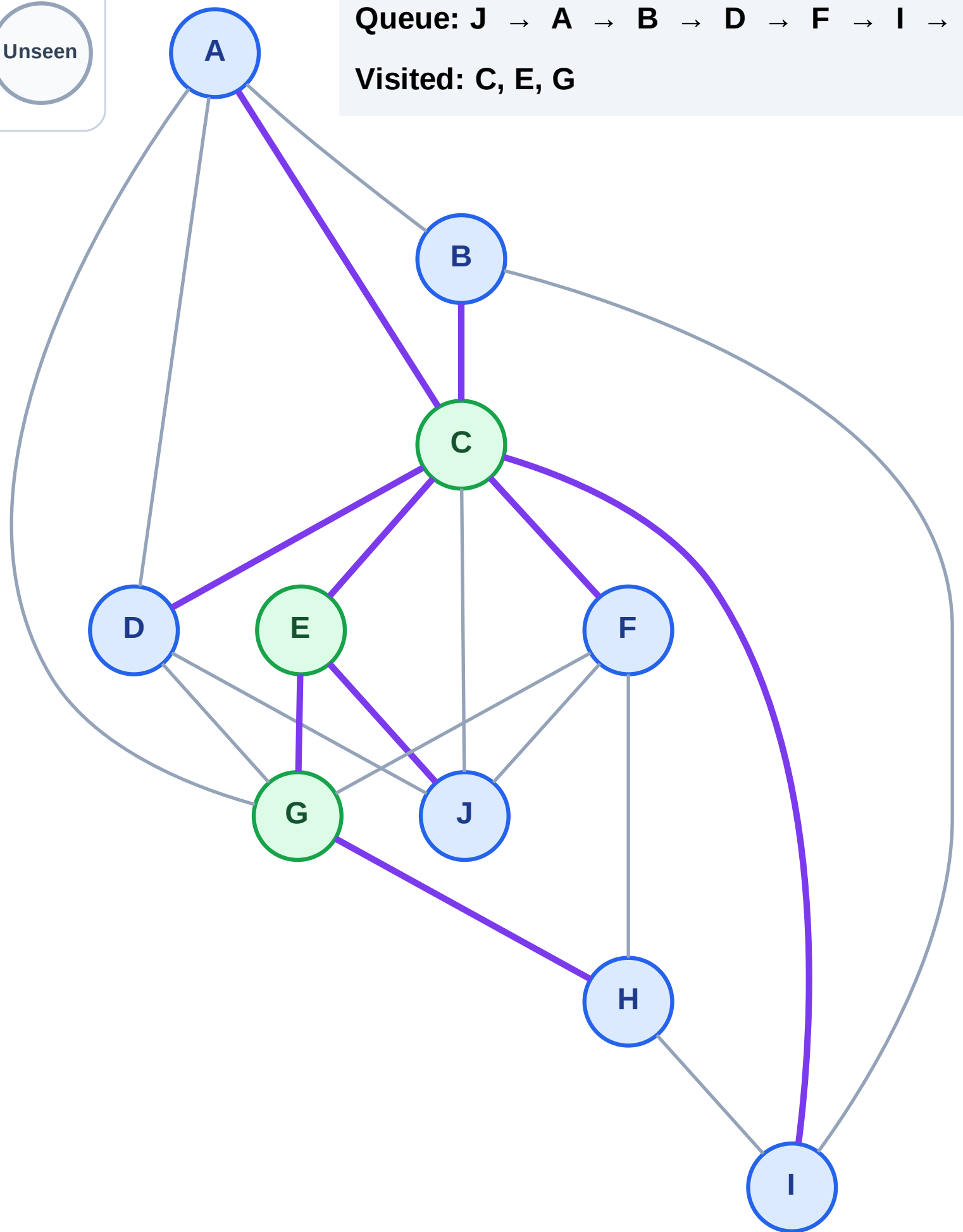
Step 7: Discover H from G

Legend



Queue: J → A → B → D → F → I → H

Visited: C, E, G



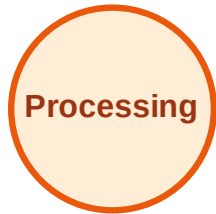
BFS Traversal

Step 8: Process node J

Legend



Complete



Processing



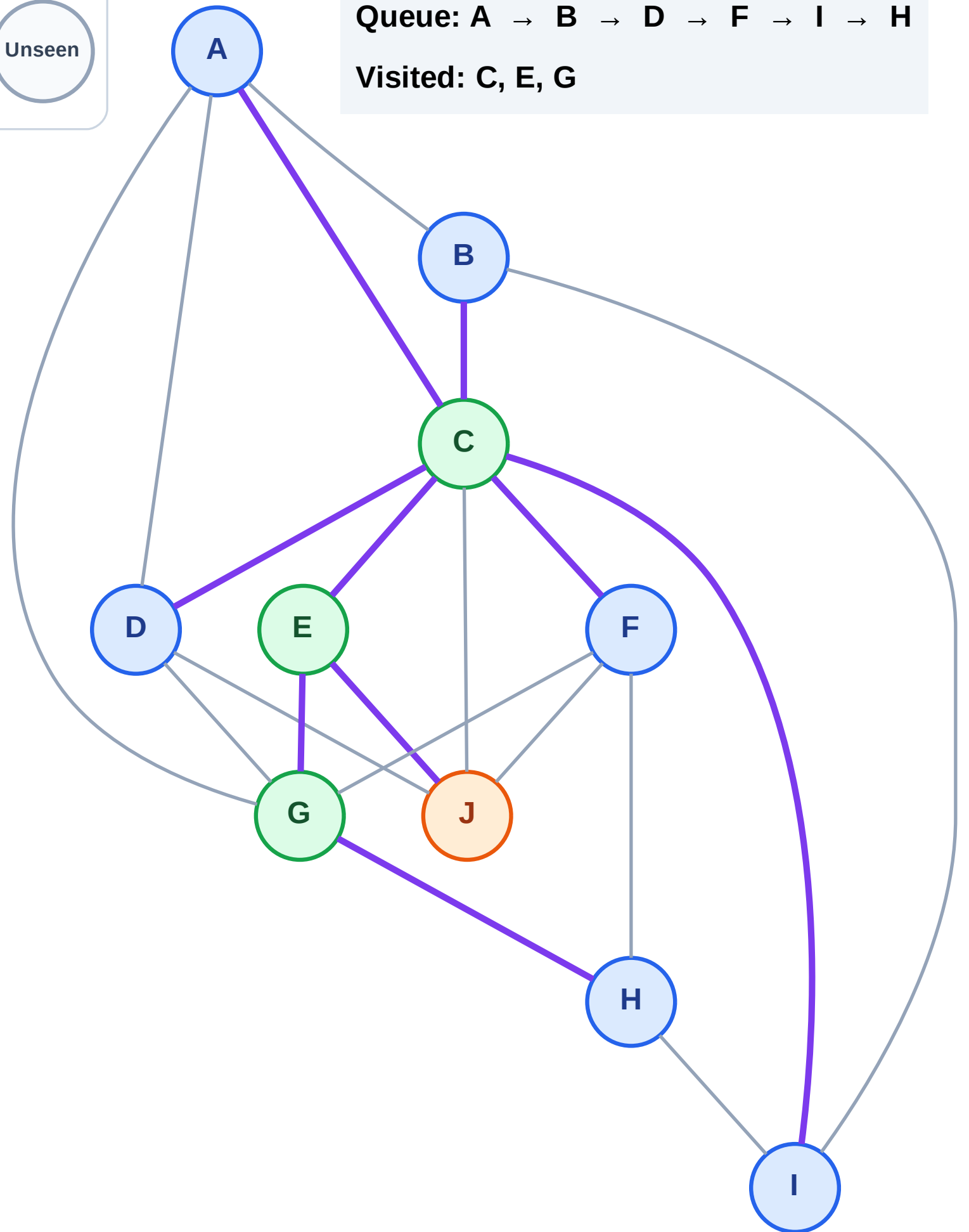
Discovered



Unseen

Queue: A → B → D → F → I → H

Visited: C, E, G



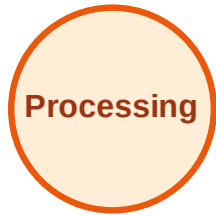
BFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing



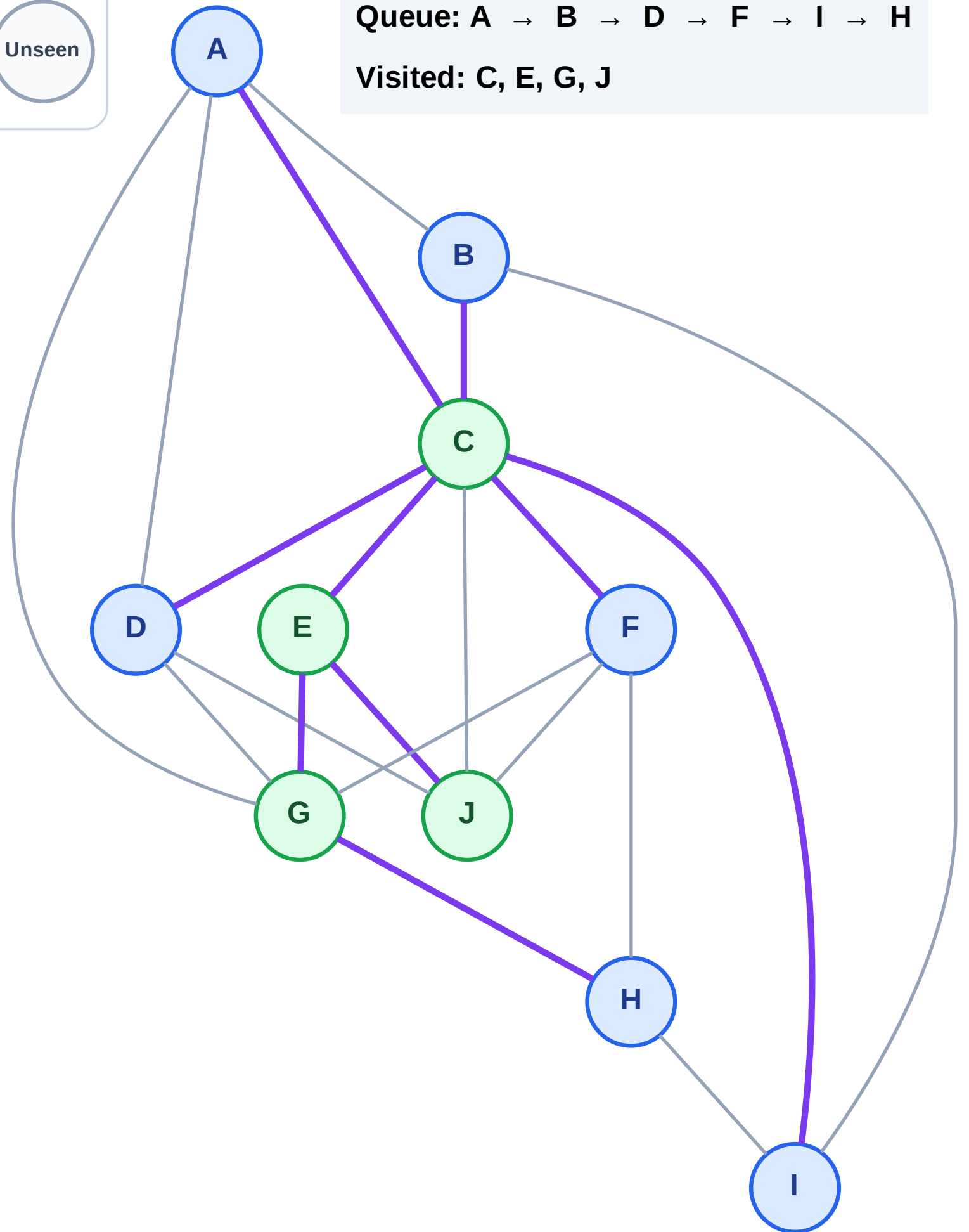
Discovered



Unseen

Queue: A → B → D → F → I → H

Visited: C, E, G, J



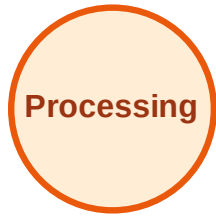
BFS Traversal

Step 10: Process node A

Legend



Complete



Processing

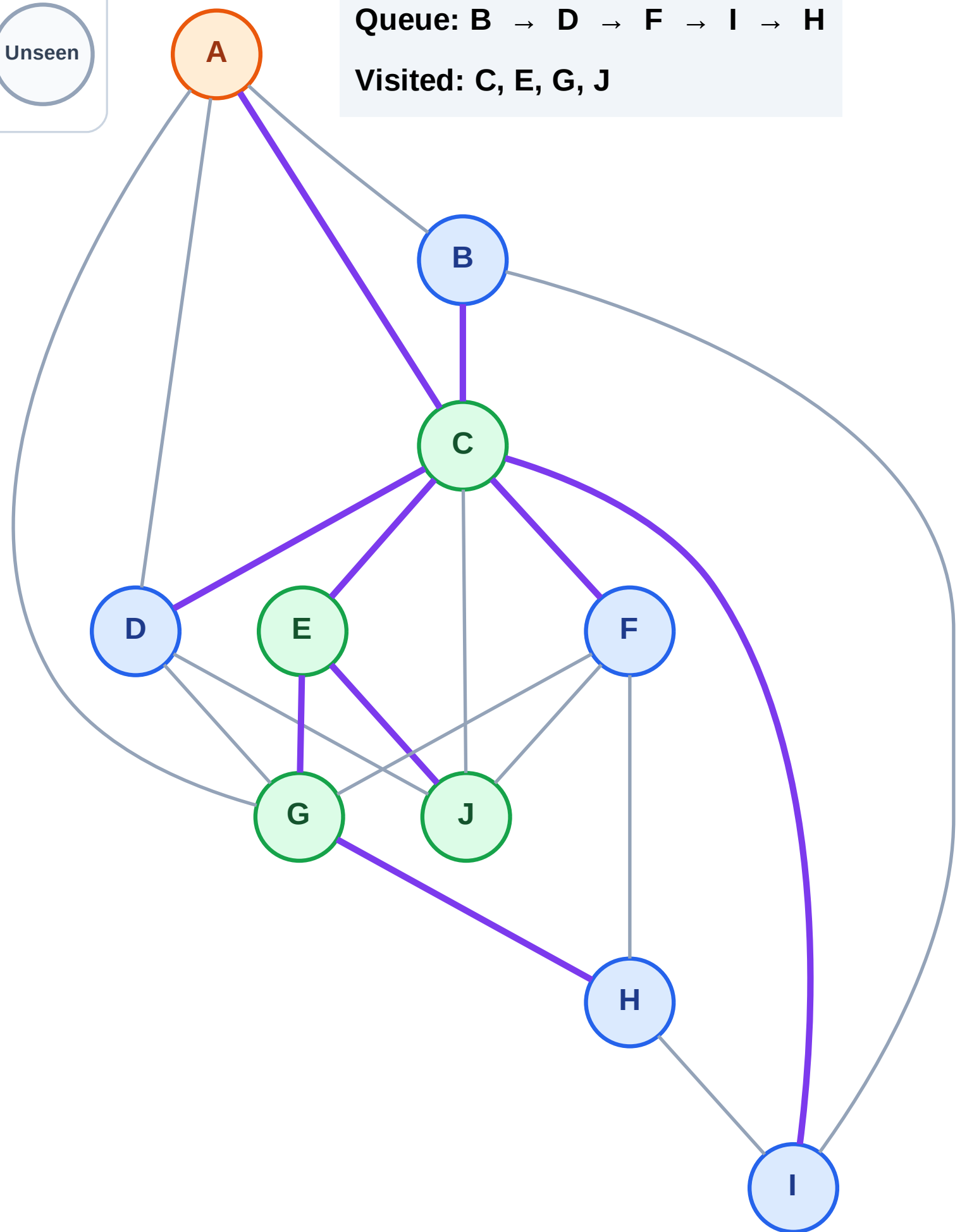


Discovered



Unseen

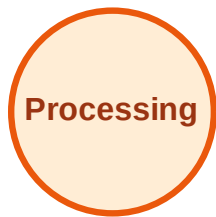
Queue: B → D → F → I → H
Visited: C, E, G, J



BFS Traversal

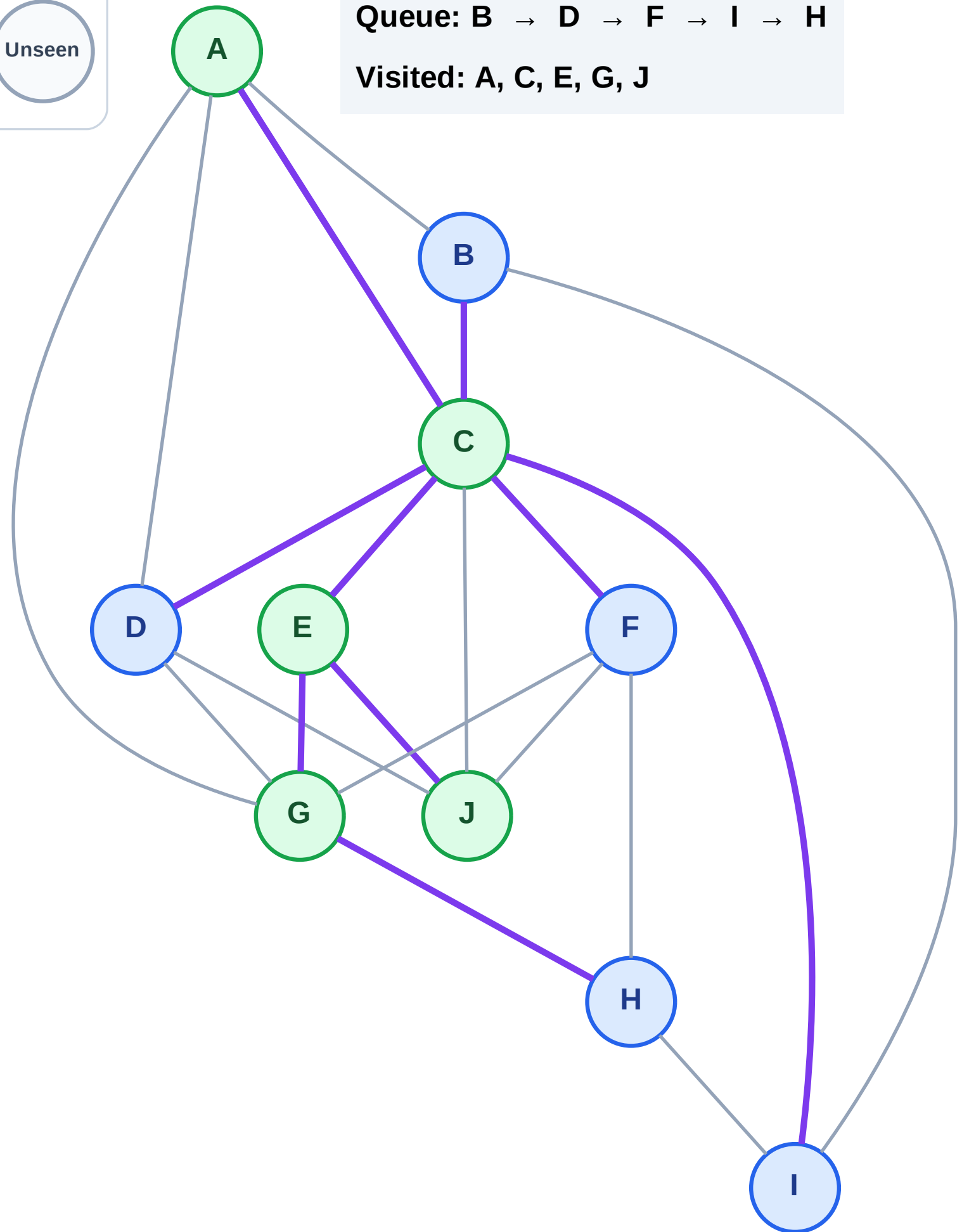
Step 11: No new neighbors from A

Legend



Queue: B → D → F → I → H

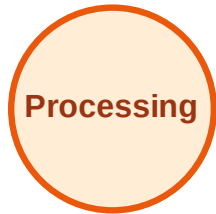
Visited: A, C, E, G, J



BFS Traversal

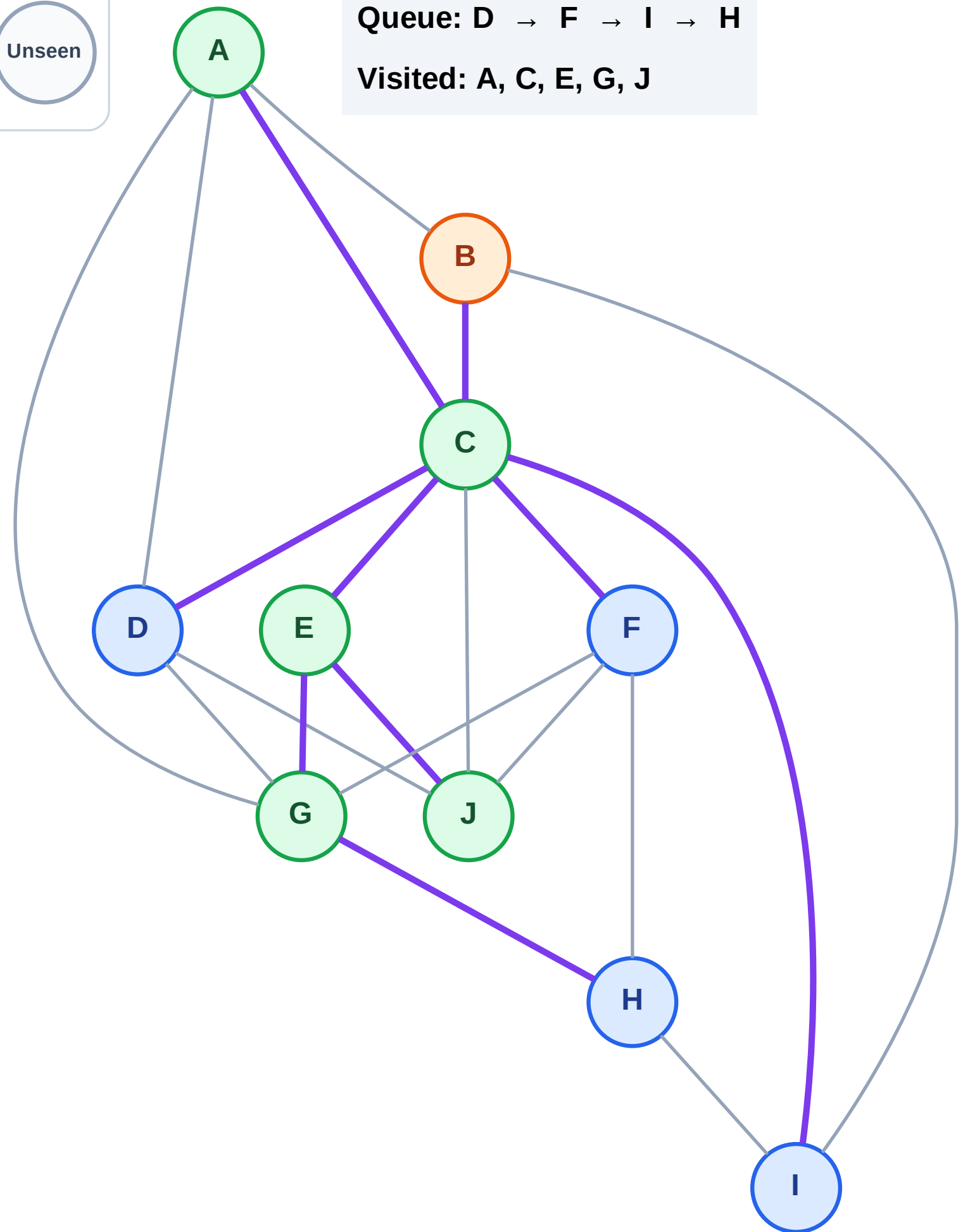
Step 12: Process node B

Legend



Queue: D → F → I → H

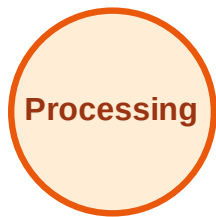
Visited: A, C, E, G, J



BFS Traversal

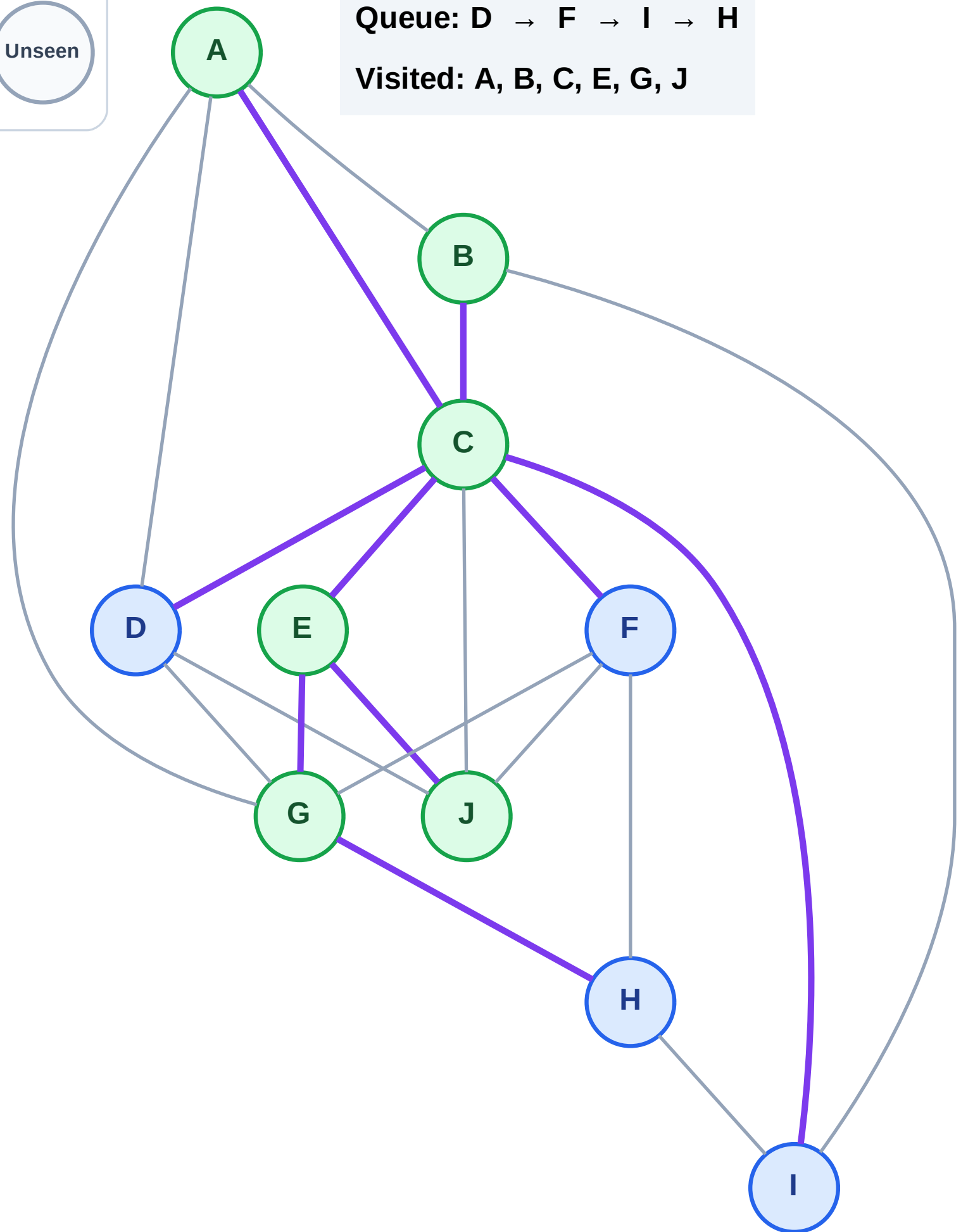
Step 13: No new neighbors from B

Legend



Queue: D → F → I → H

Visited: A, B, C, E, G, J



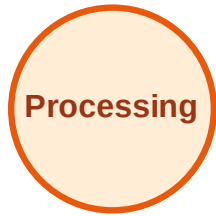
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



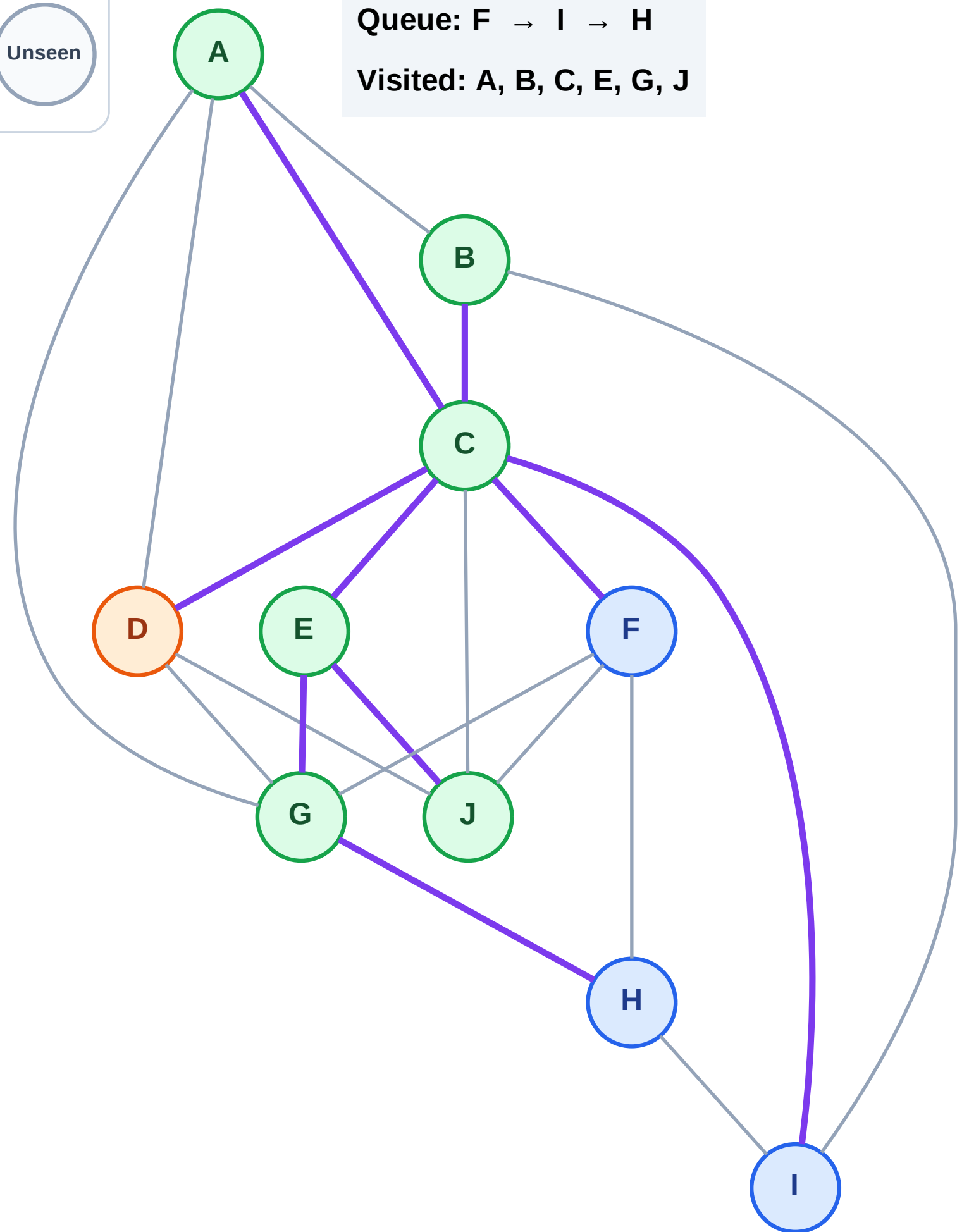
Discovered



Unseen

Queue: F → I → H

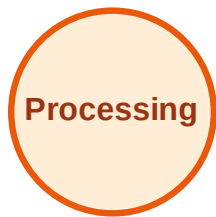
Visited: A, B, C, E, G, J



BFS Traversal

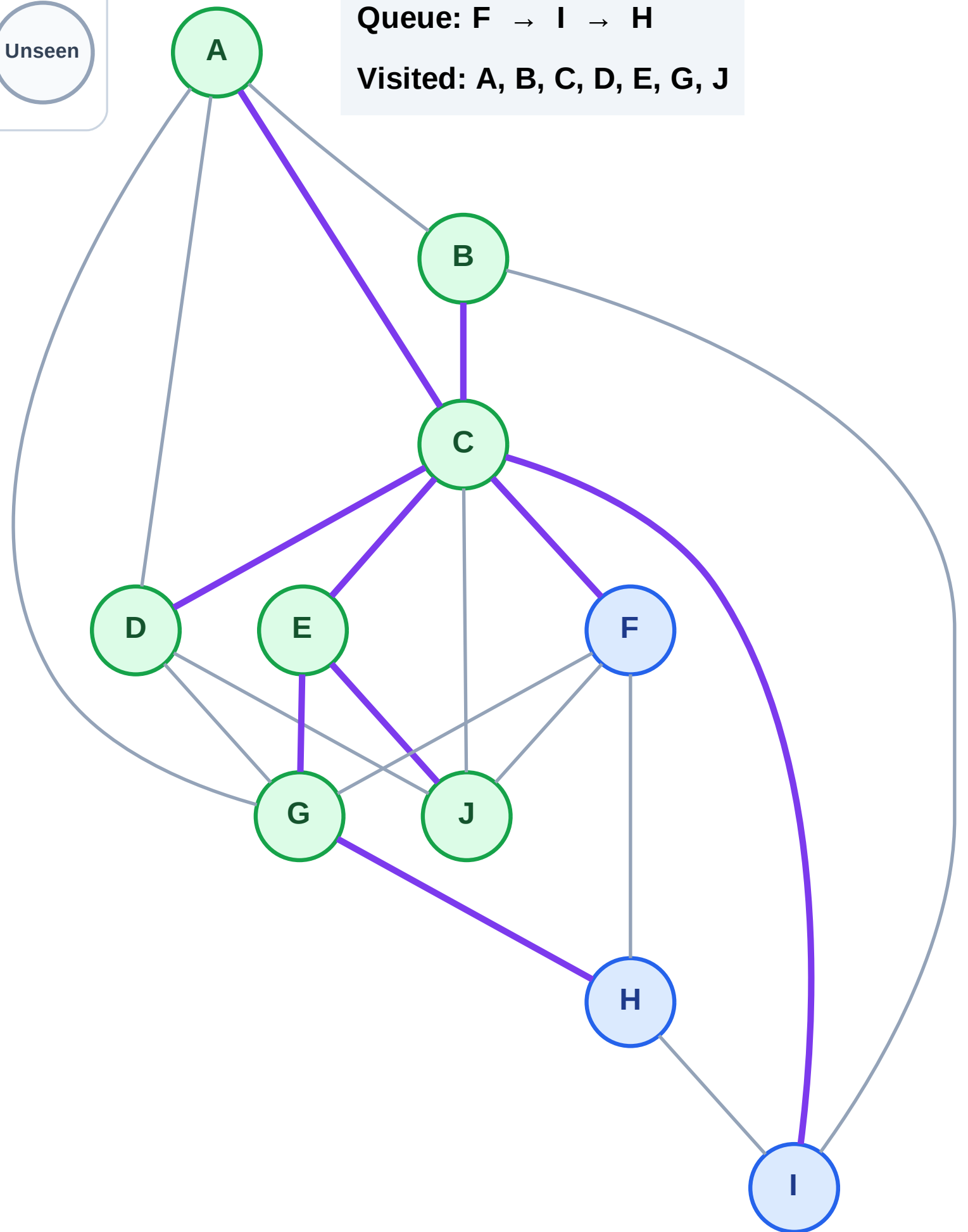
Step 15: No new neighbors from D

Legend



Queue: F → I → H

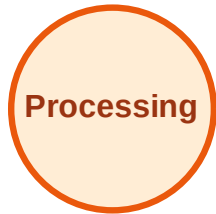
Visited: A, B, C, D, E, G, J



BFS Traversal

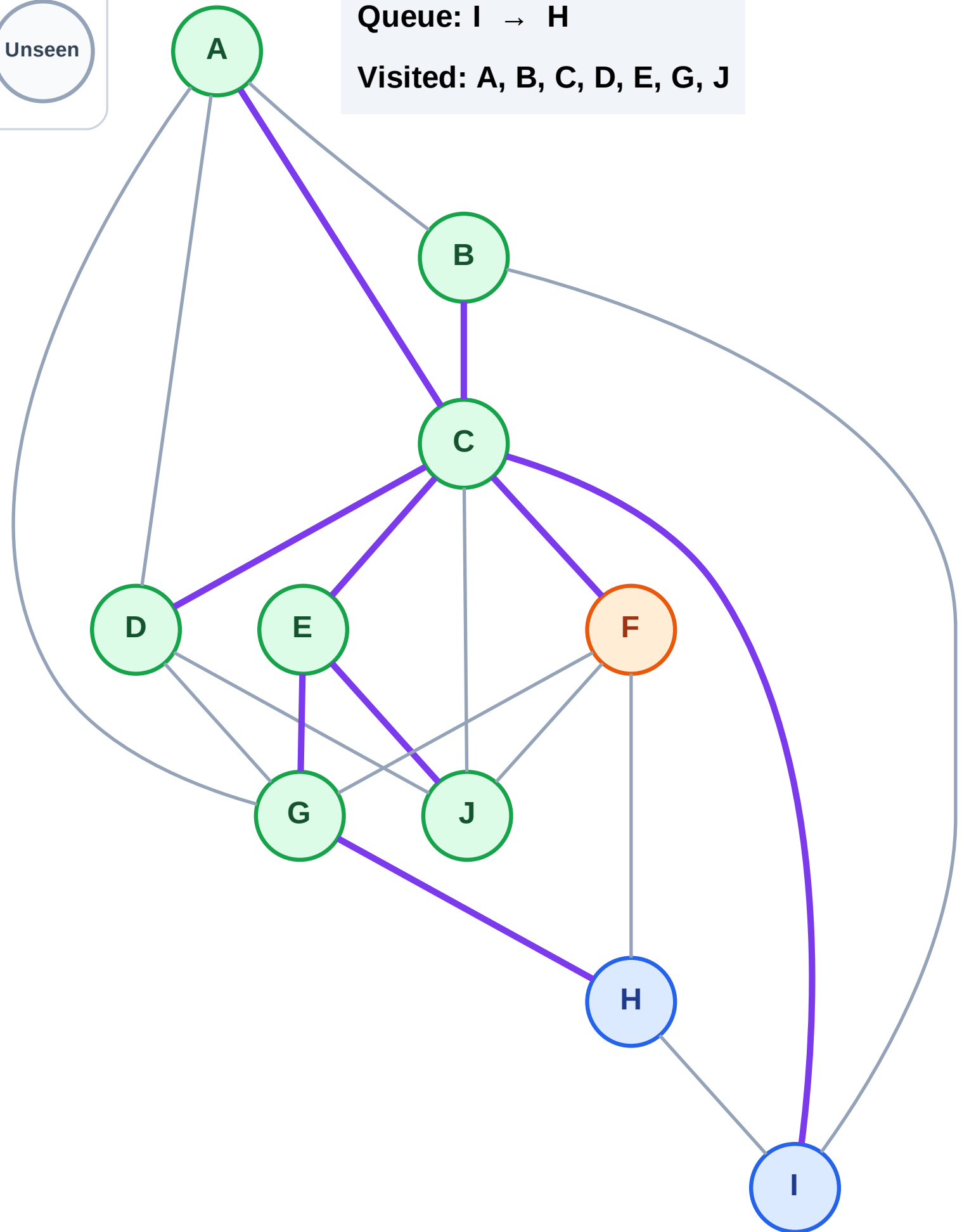
Step 16: Process node F

Legend



Queue: I → H

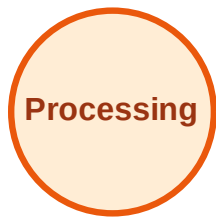
Visited: A, B, C, D, E, G, J



BFS Traversal

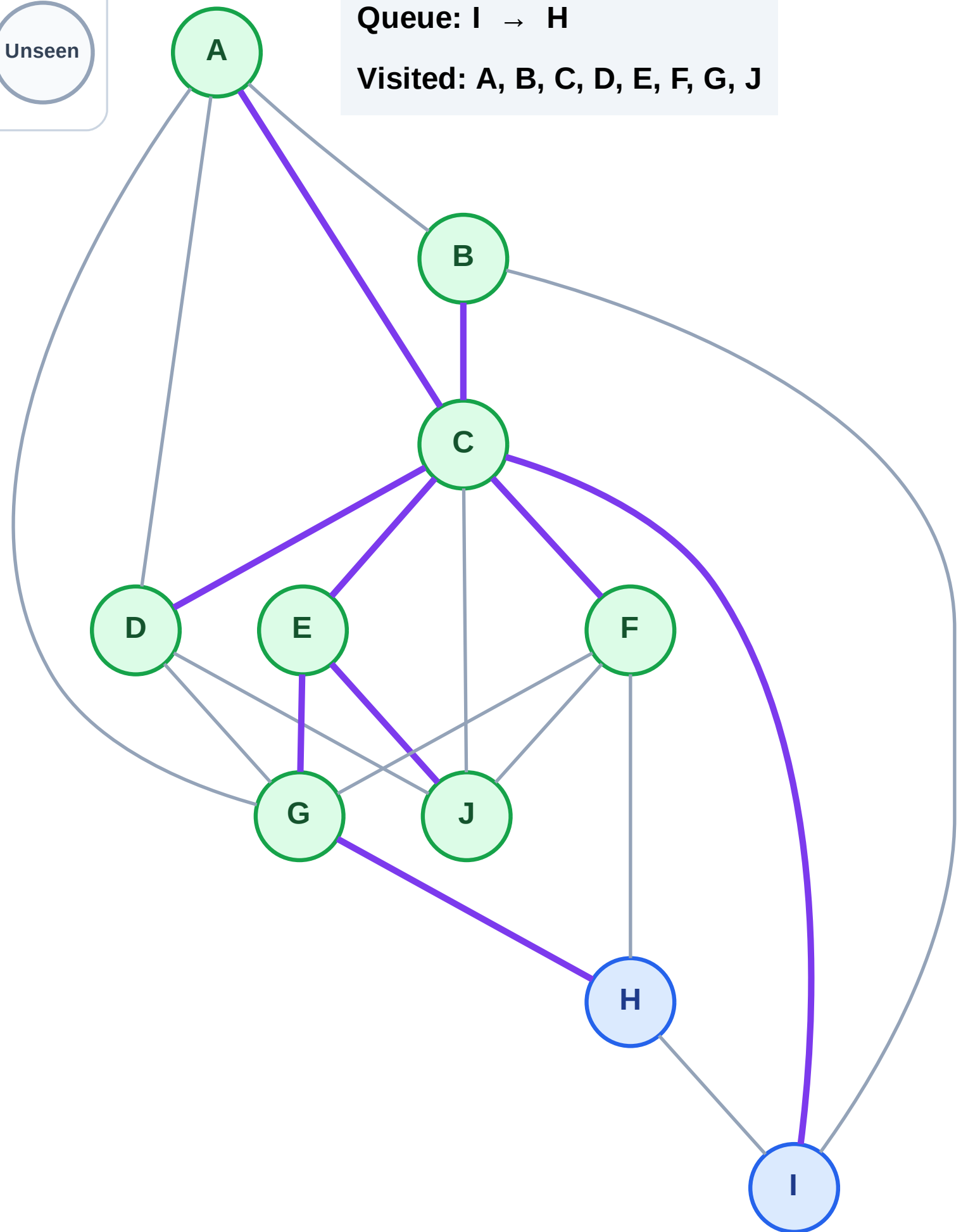
Step 17: No new neighbors from F

Legend



Queue: I → H

Visited: A, B, C, D, E, F, G, J



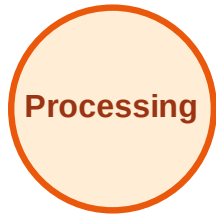
BFS Traversal

Step 18: Process node I

Legend



Complete



Processing



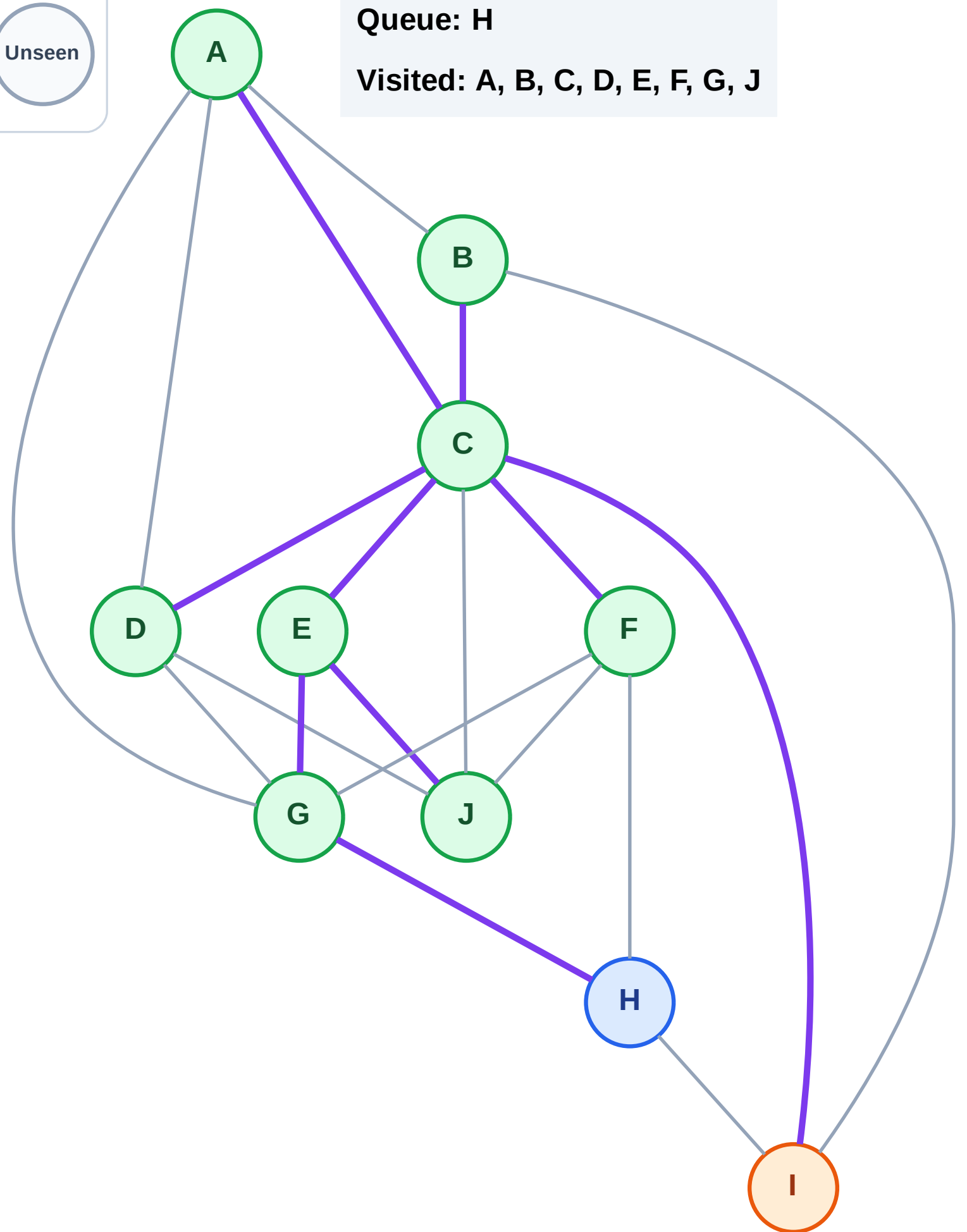
Discovered



Unseen

Queue: H

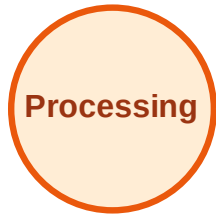
Visited: A, B, C, D, E, F, G, J



BFS Traversal

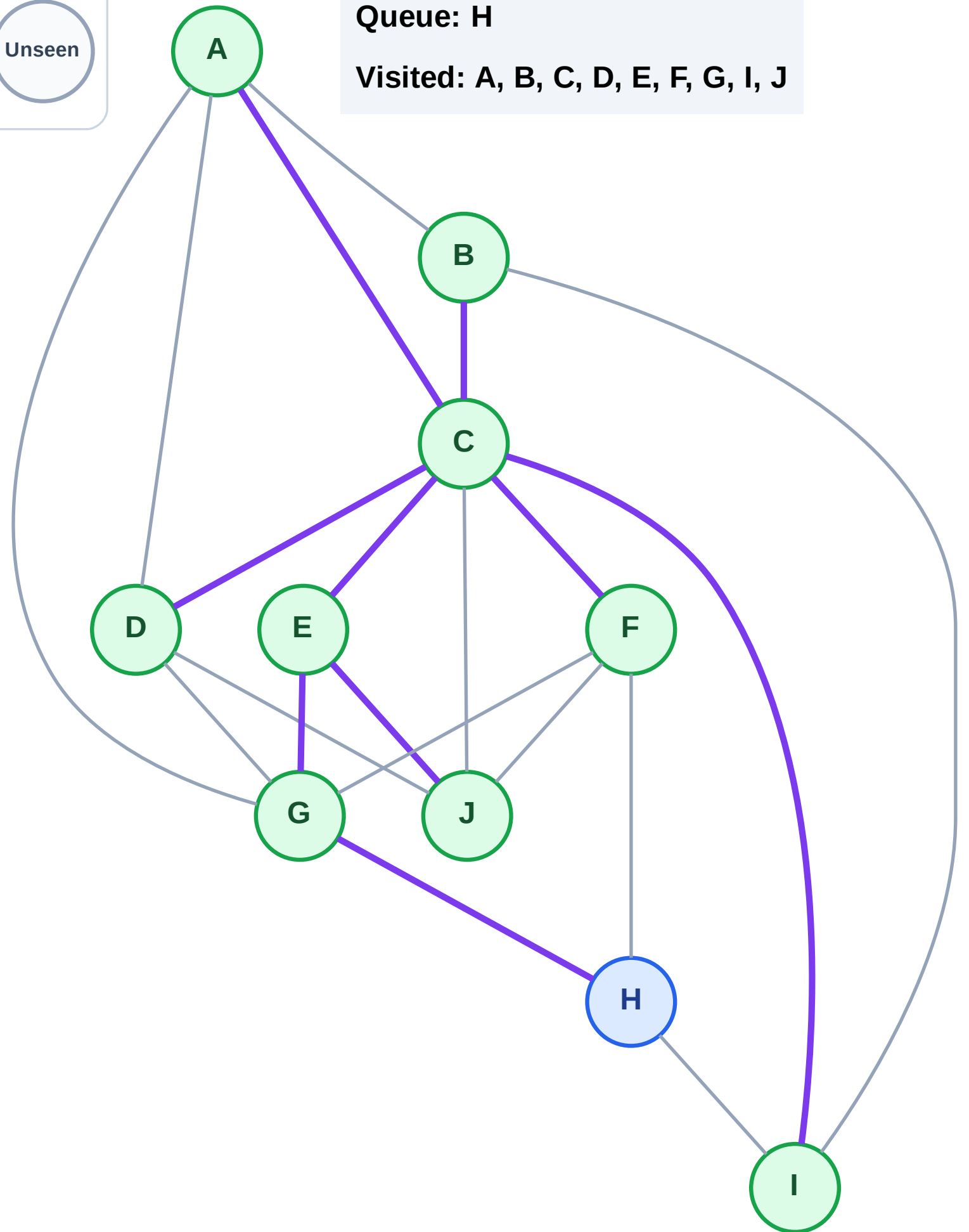
Step 19: No new neighbors from I

Legend



Queue: H

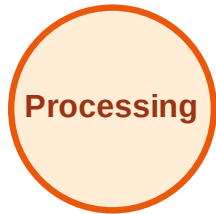
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

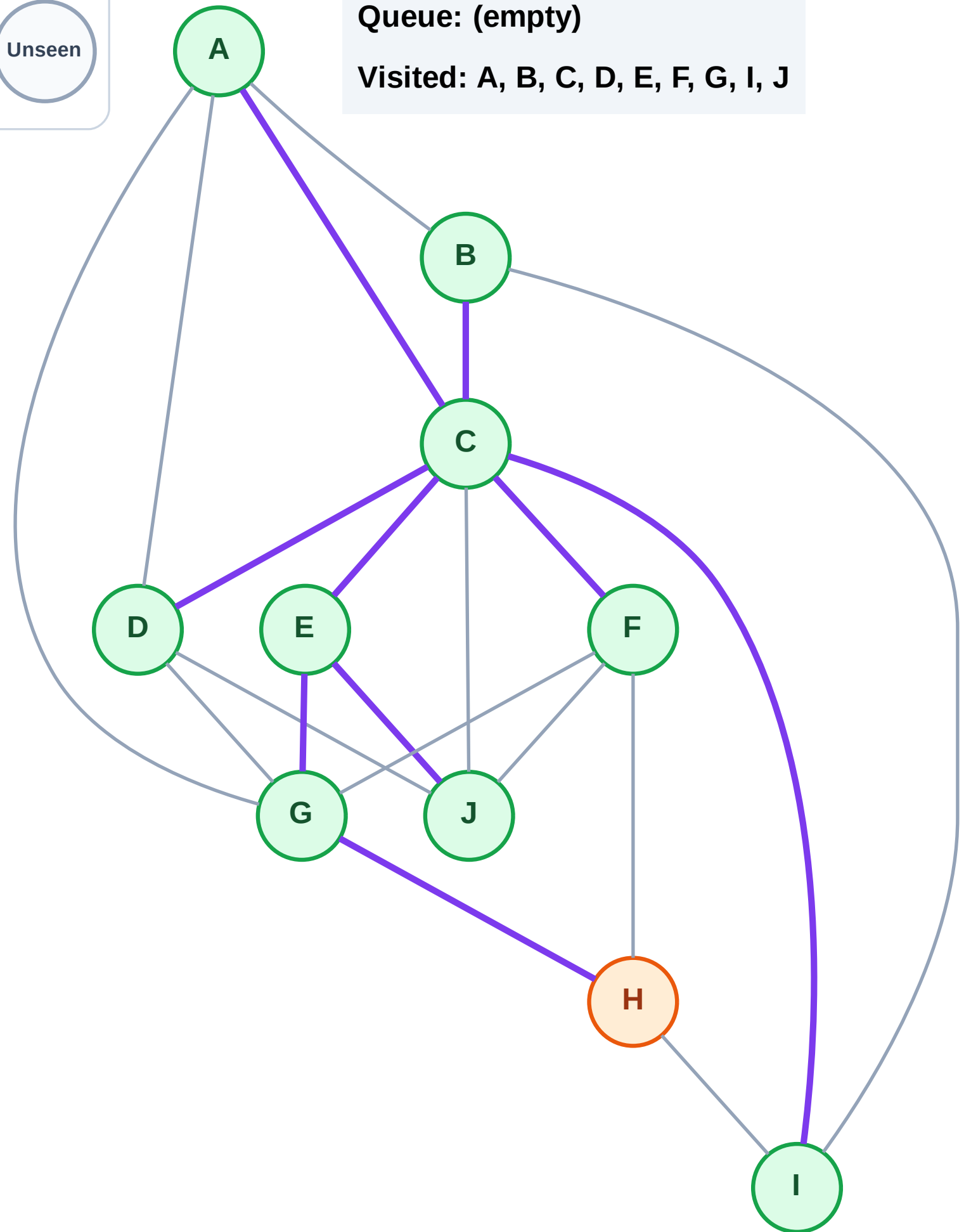
Step 20: Process node H

Legend



Queue: (empty)

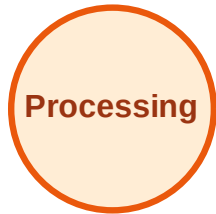
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

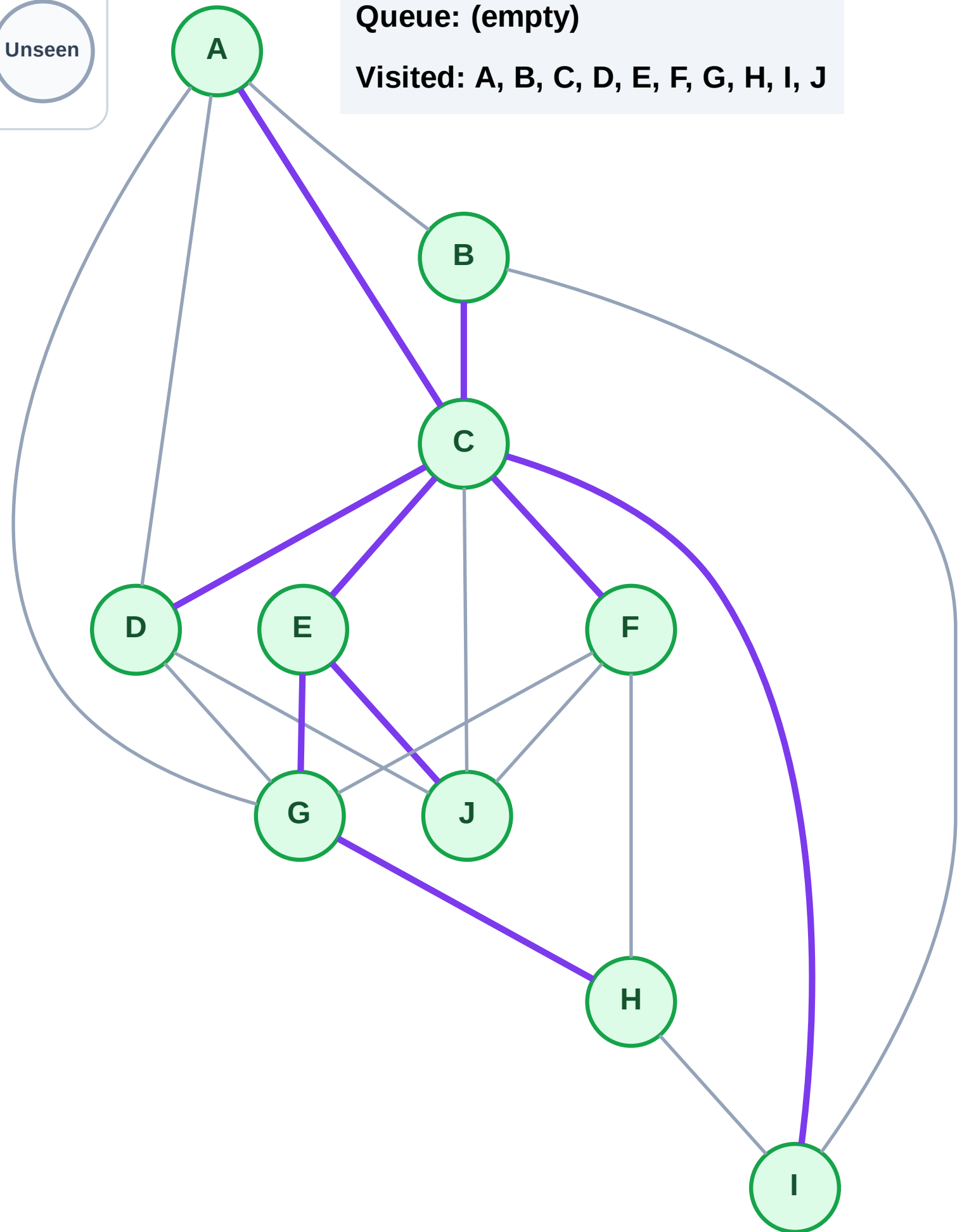
Step 21: No new neighbors from H

Legend



Queue: (empty)

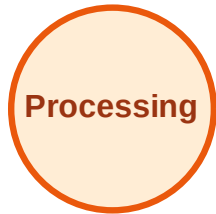
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

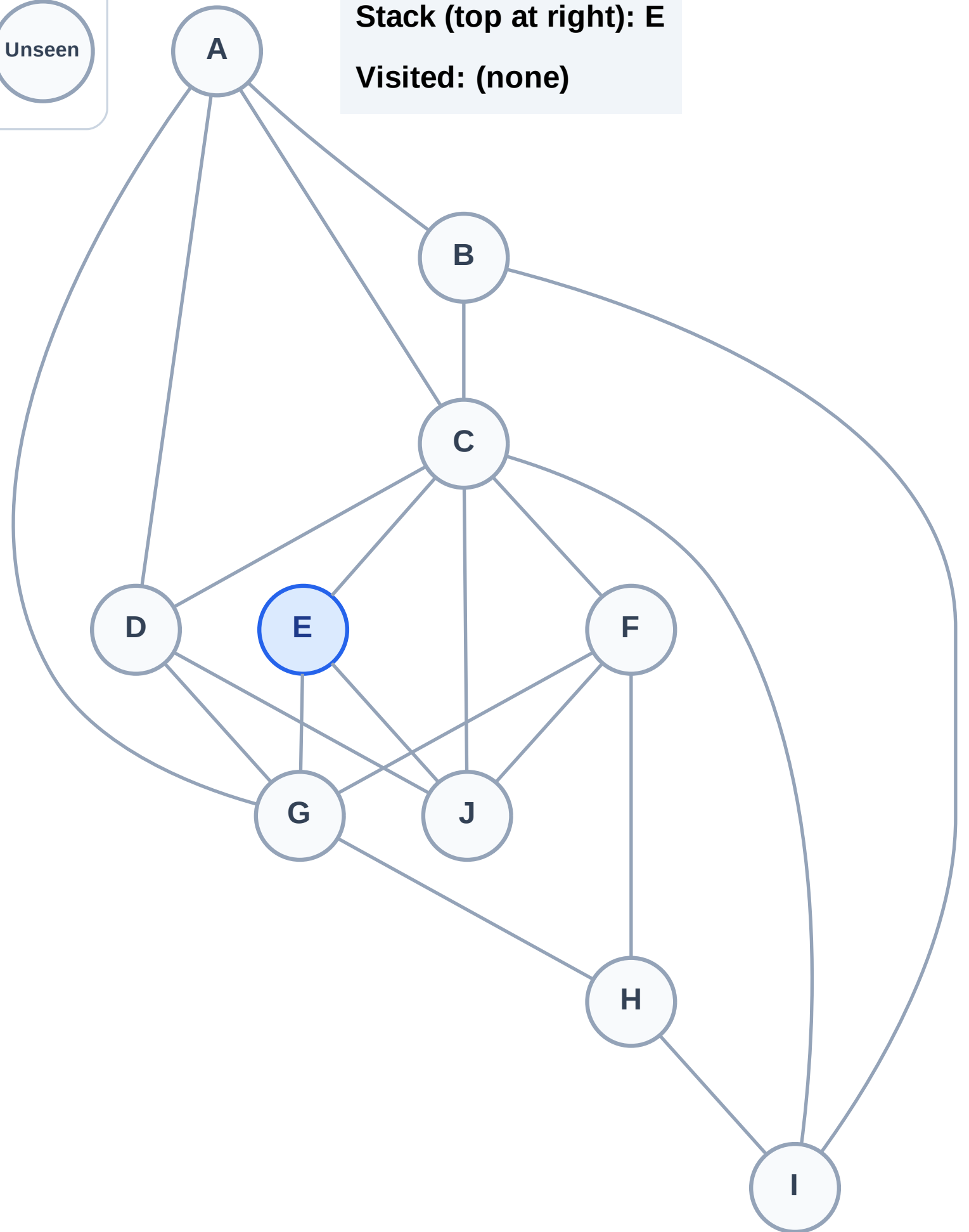
Step 1: Start at E

Legend



Stack (top at right): E

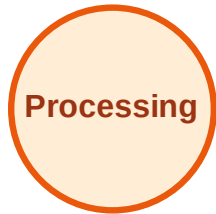
Visited: (none)



DFS Traversal

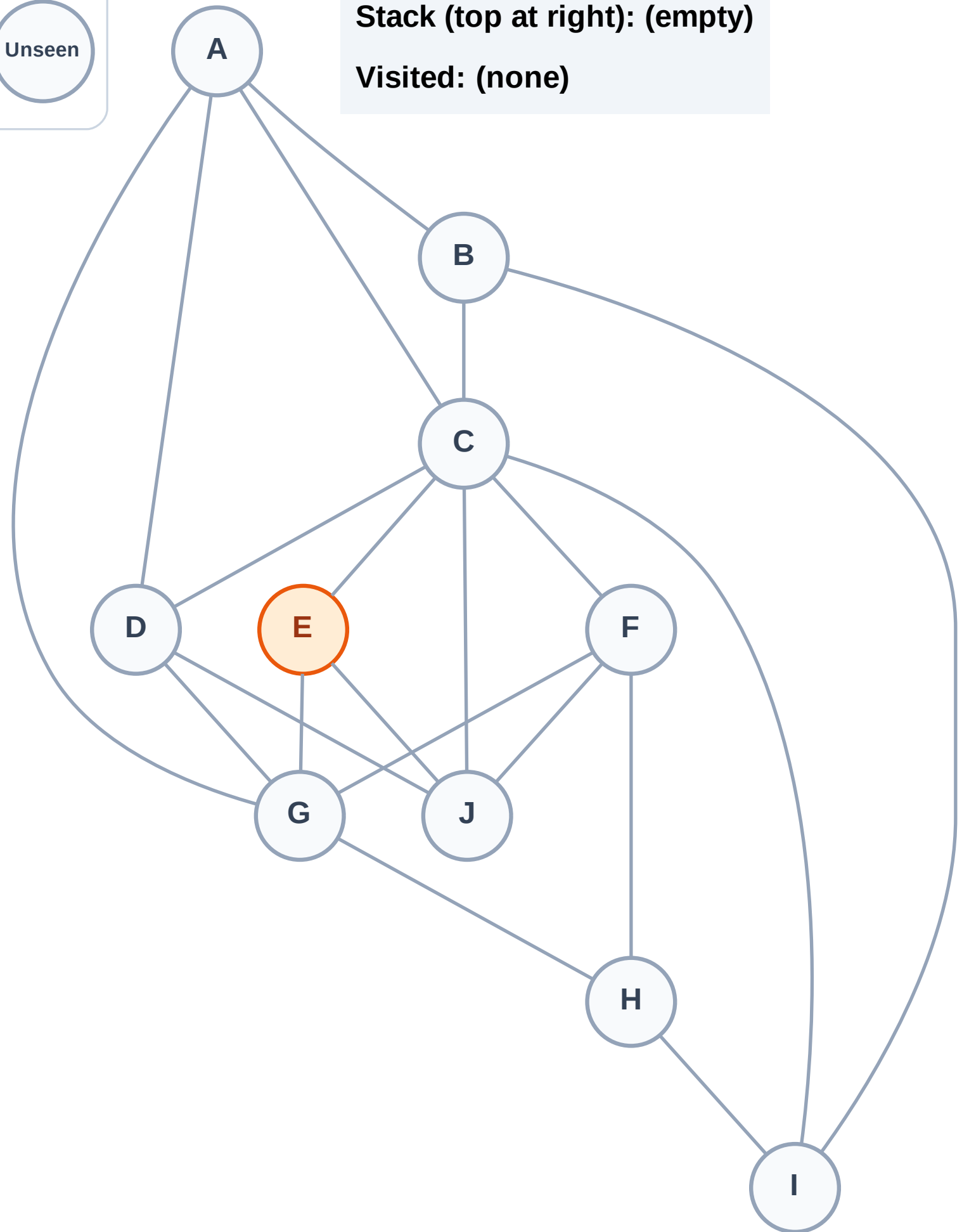
Step 2: Process node E

Legend



Stack (top at right): (empty)

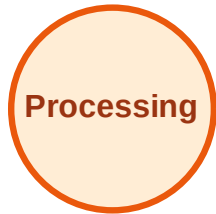
Visited: (none)



DFS Traversal

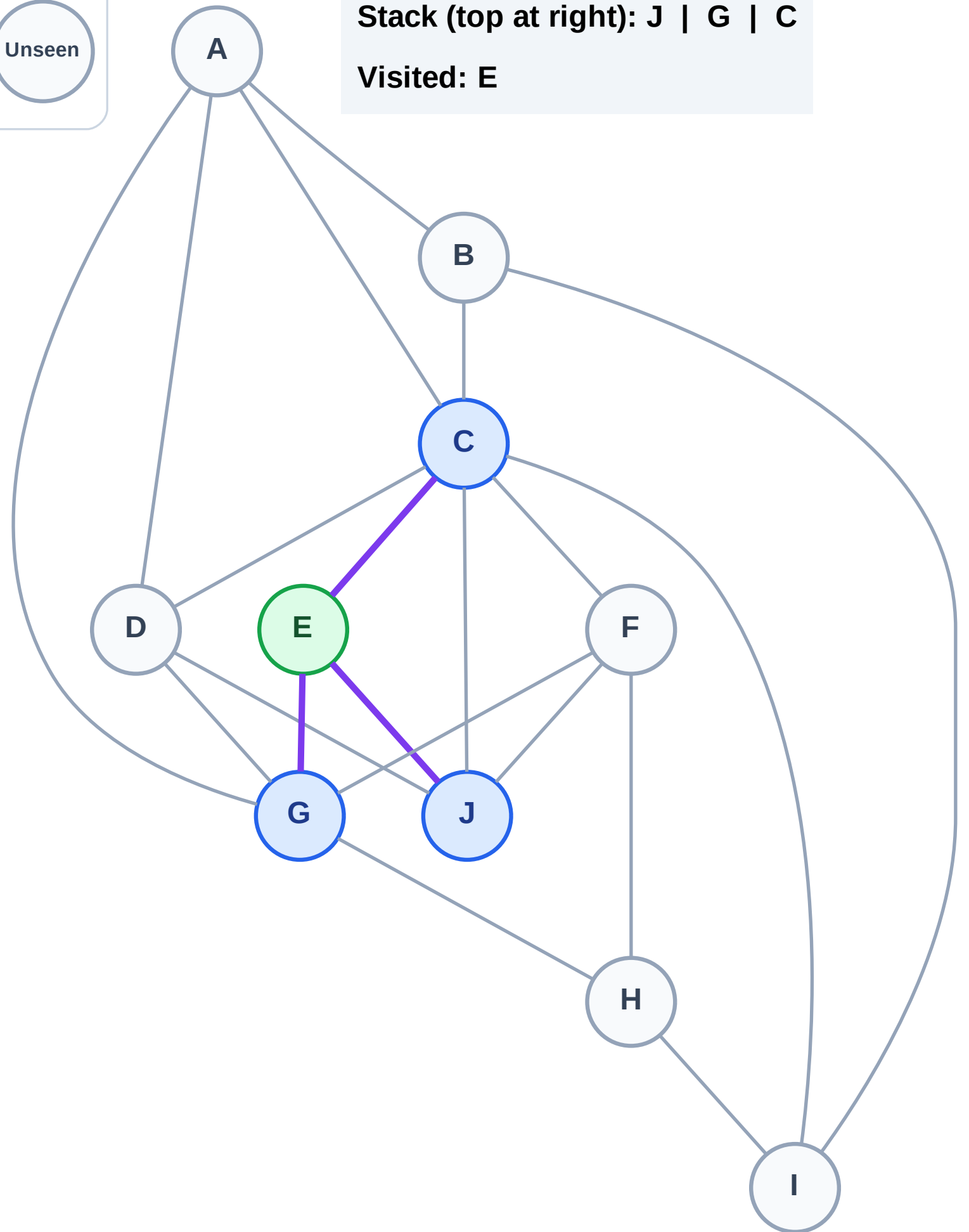
Step 3: Discover J, G, C from E

Legend



Stack (top at right): J | G | C

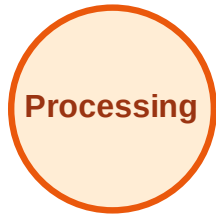
Visited: E



DFS Traversal

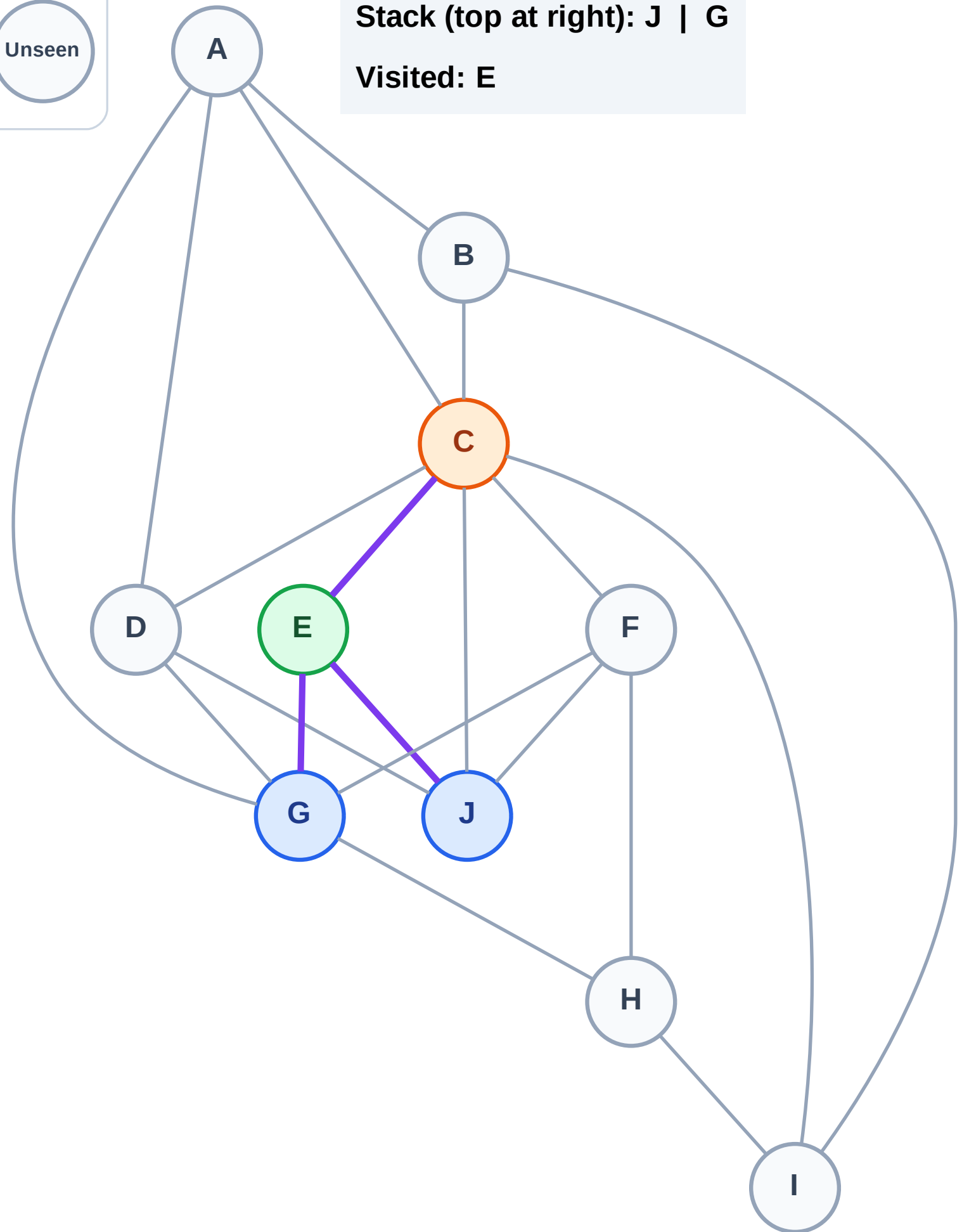
Step 4: Process node C

Legend



Stack (top at right): J | G

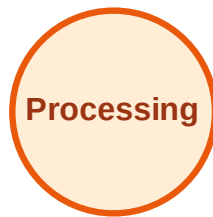
Visited: E



DFS Traversal

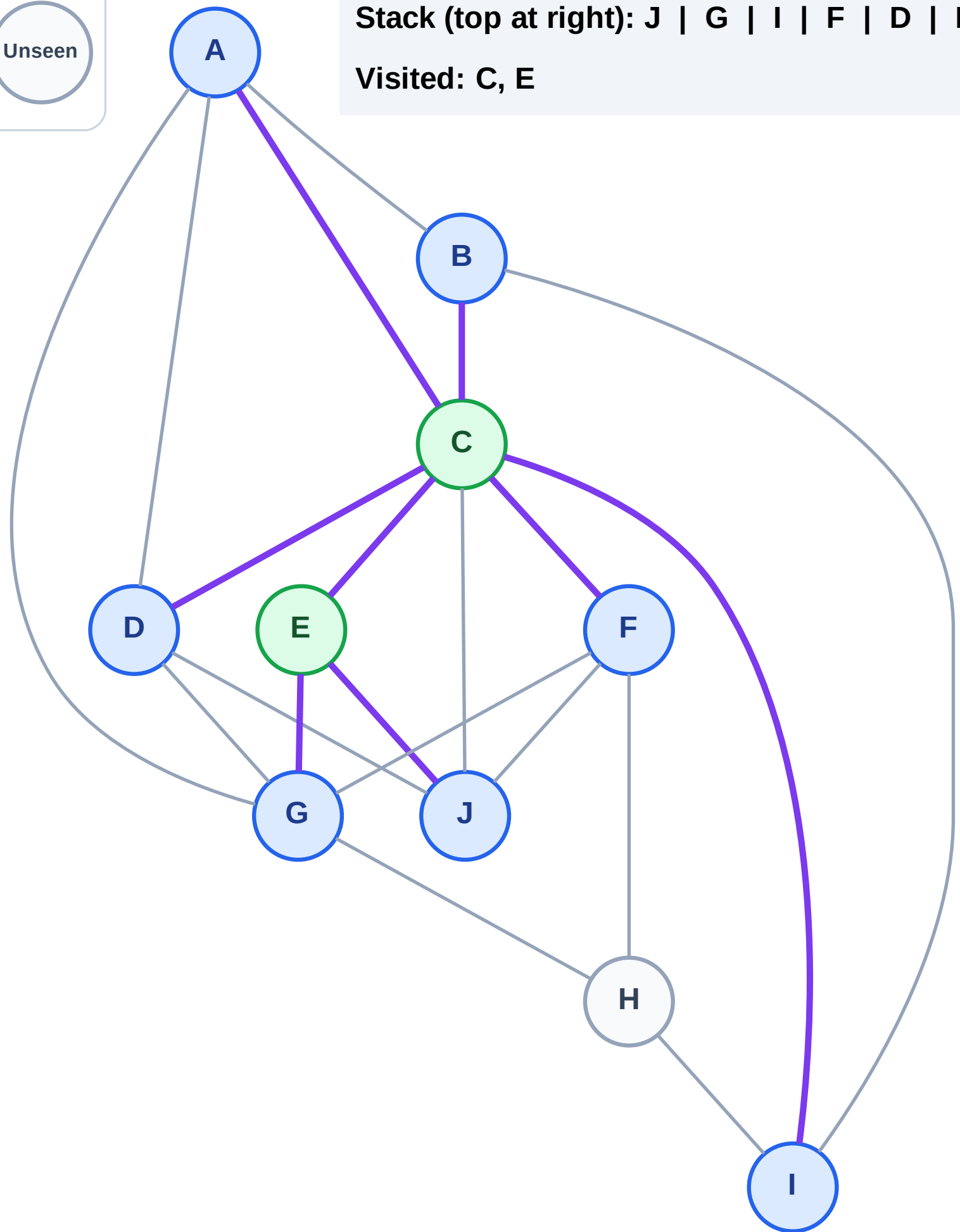
Step 5: Discover I, F, D, B, A from C

Legend



Stack (top at right): J | G | I | F | D | B | A

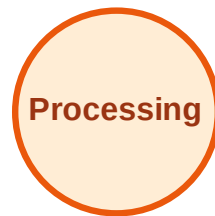
Visited: C, E



DFS Traversal

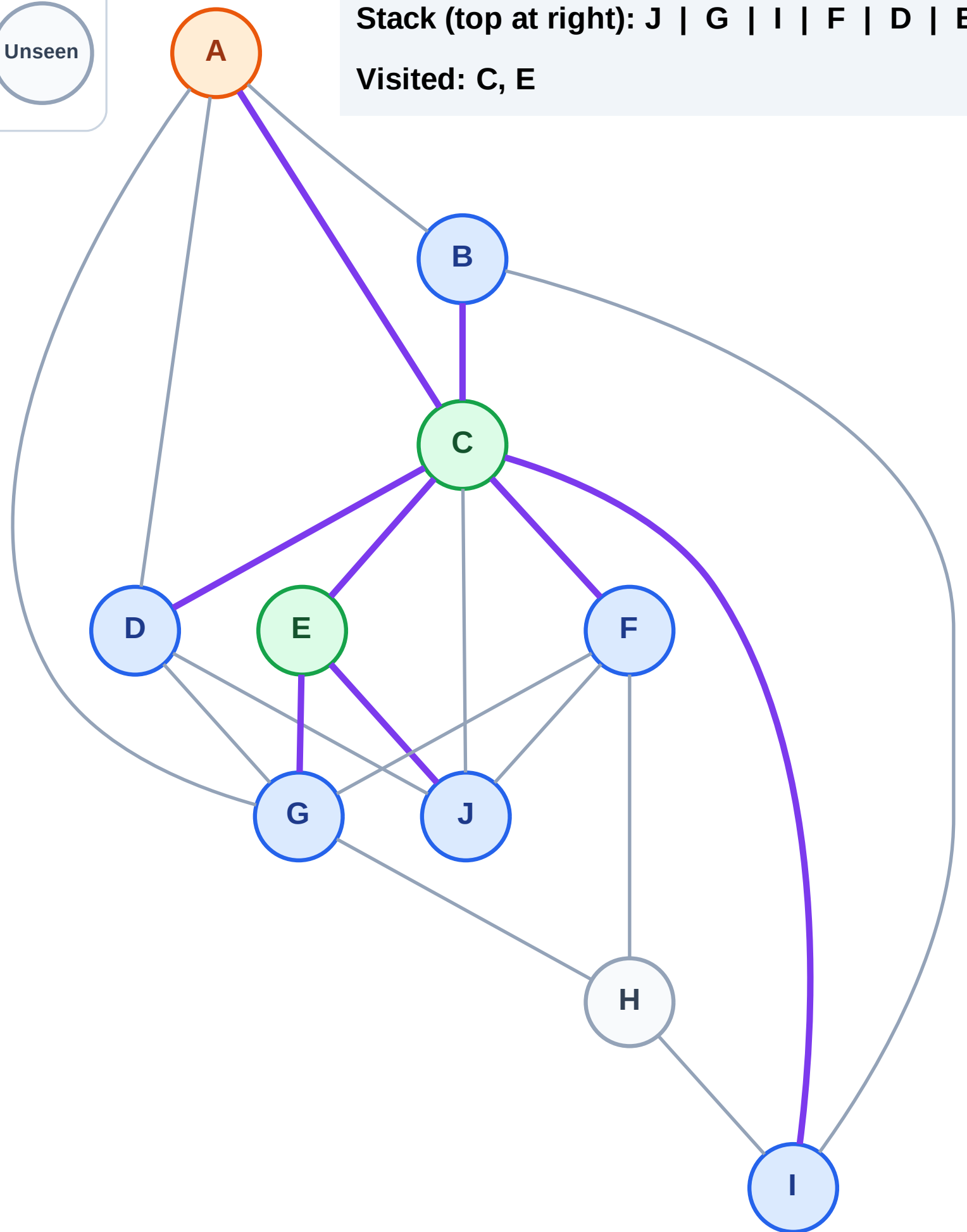
Step 6: Process node A

Legend



Stack (top at right): J | G | I | F | D | B

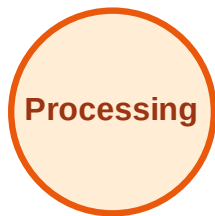
Visited: C, E



DFS Traversal

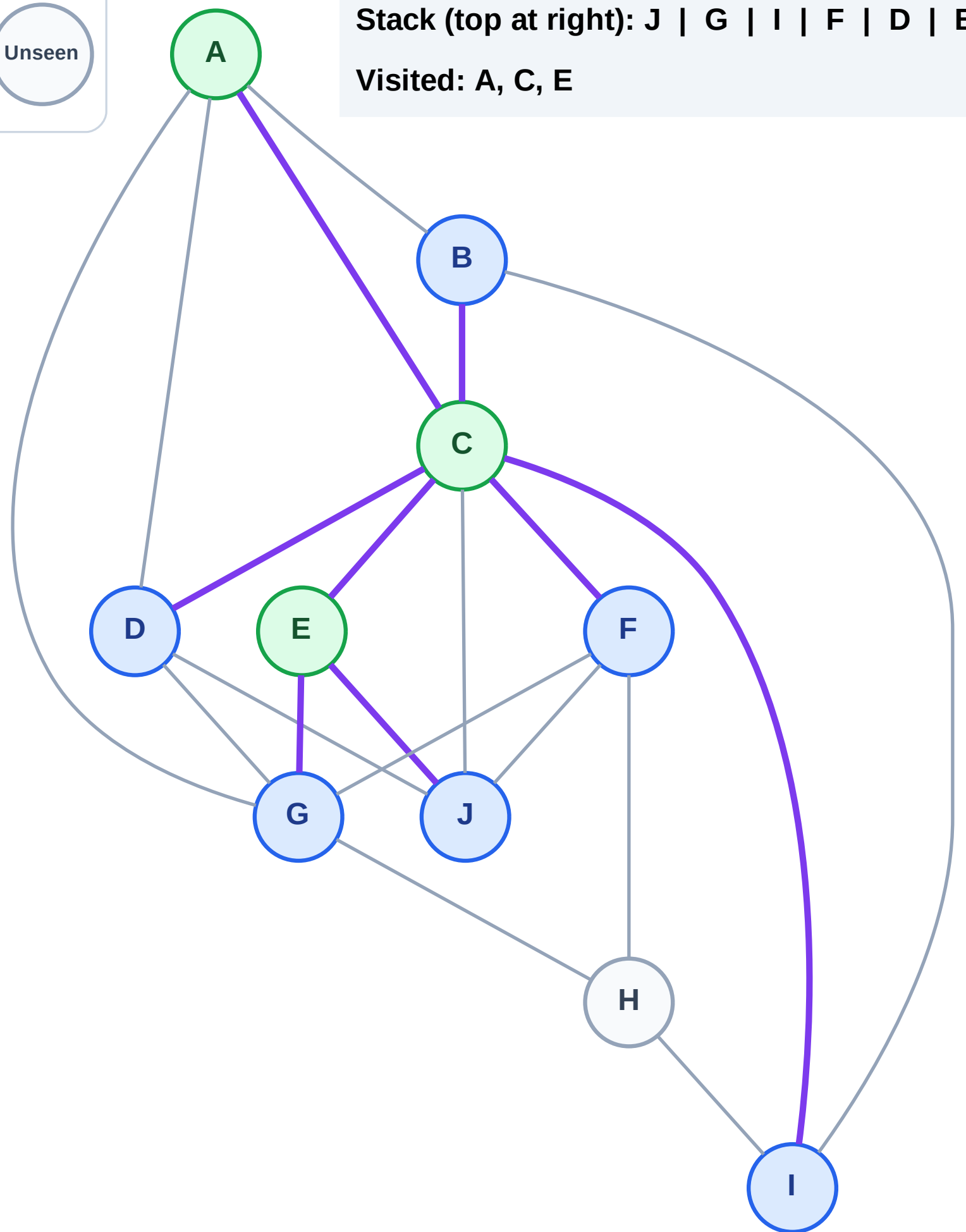
Step 7: No new neighbors from A

Legend



Stack (top at right): J | G | I | F | D | B

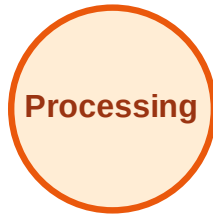
Visited: A, C, E



DFS Traversal

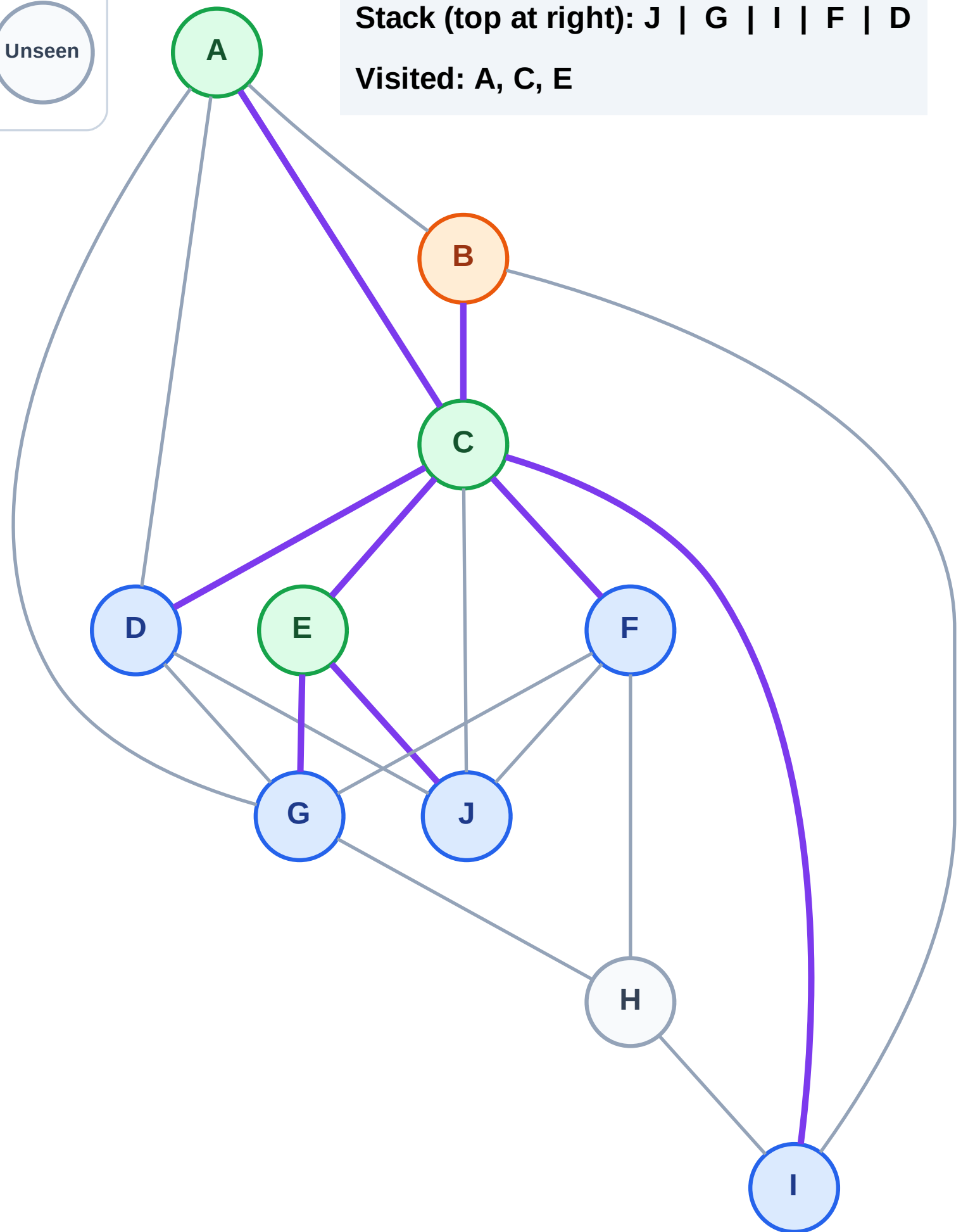
Step 8: Process node B

Legend



Stack (top at right): J | G | I | F | D

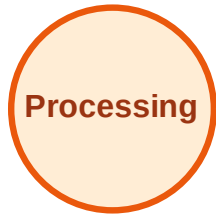
Visited: A, C, E



DFS Traversal

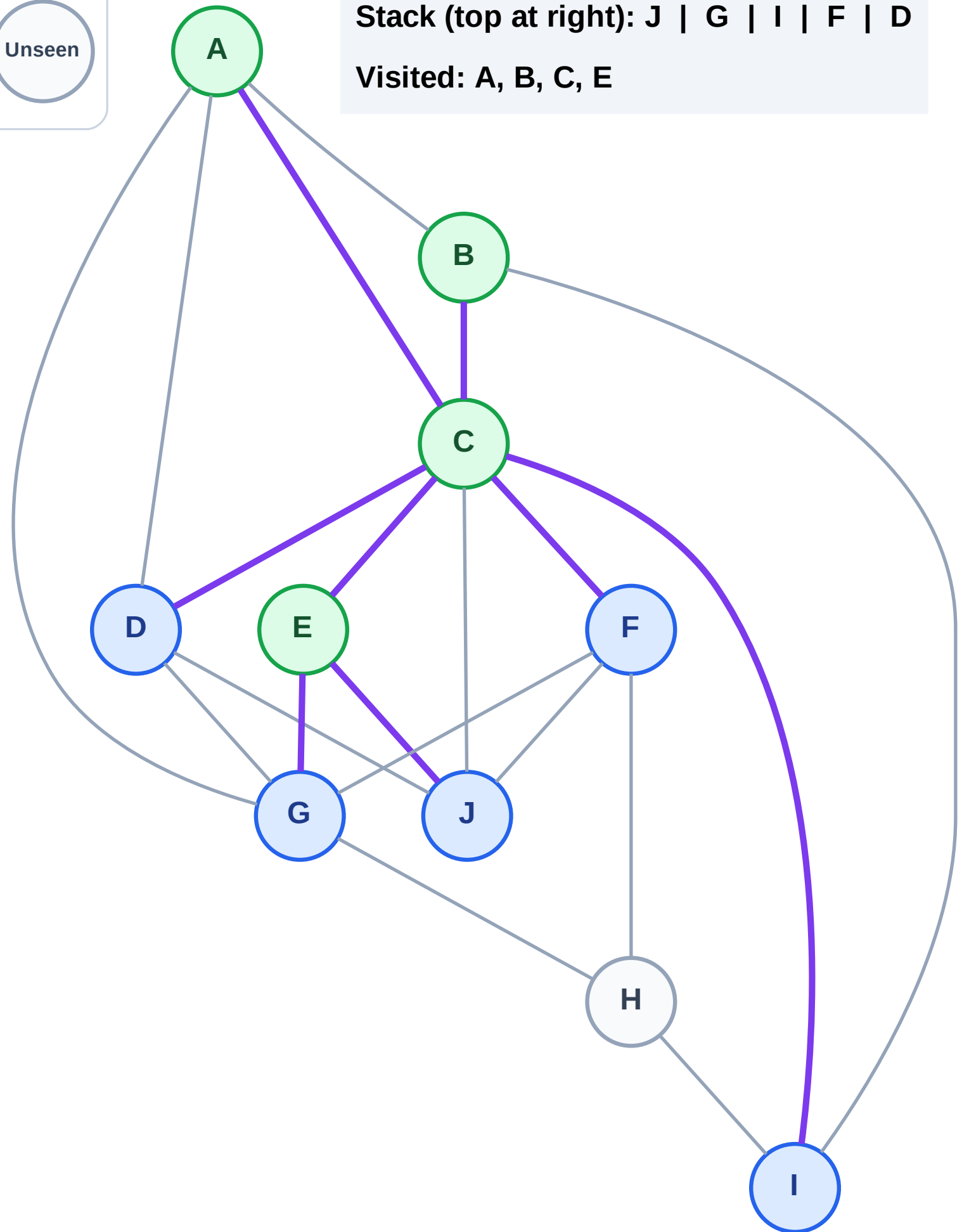
Step 9: No new neighbors from B

Legend



Stack (top at right): J | G | I | F | D

Visited: A, B, C, E



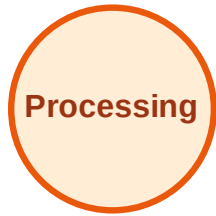
DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



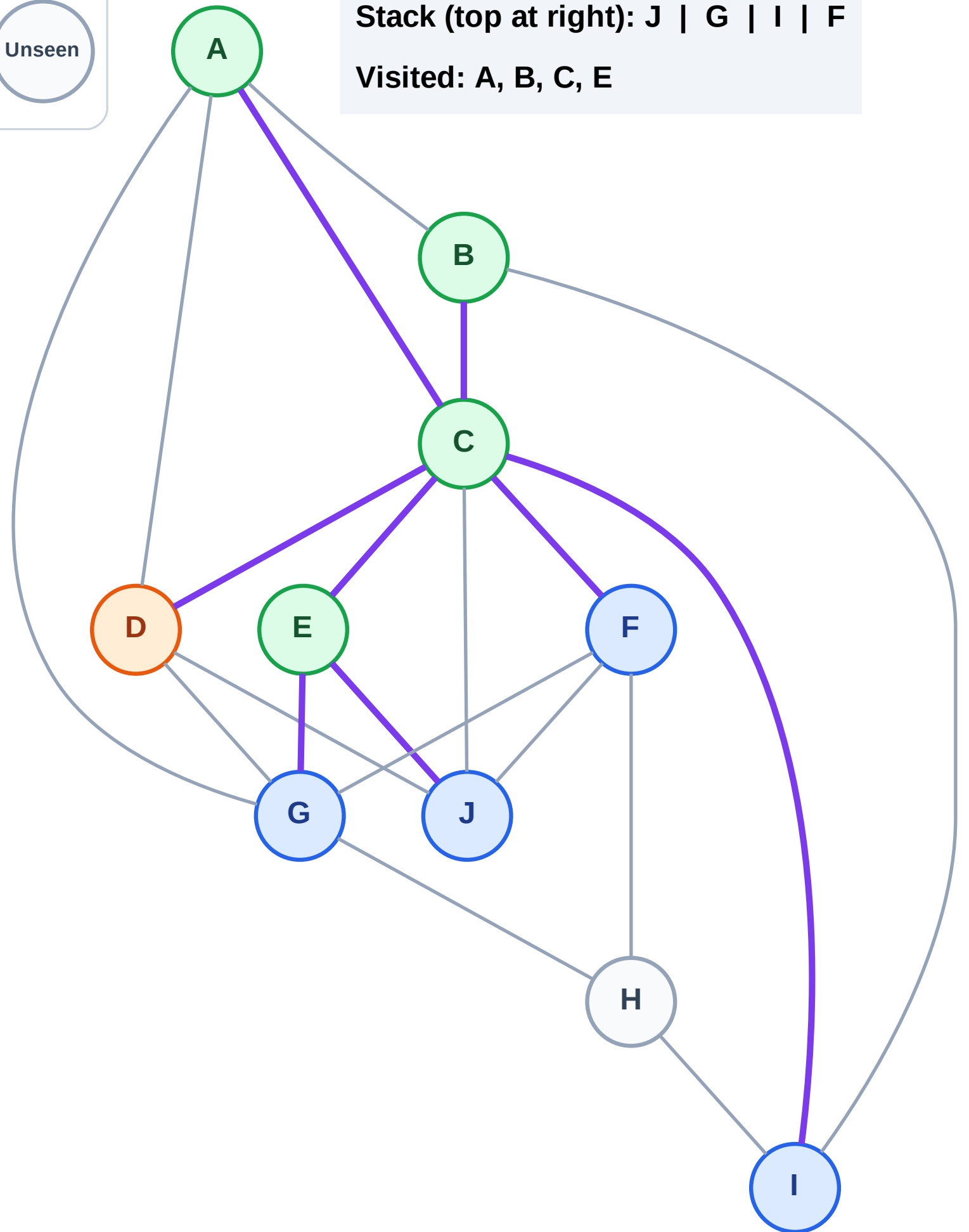
Discovered



Unseen

Stack (top at right): J | G | I | F

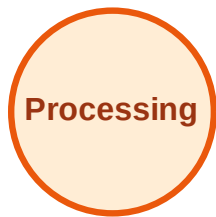
Visited: A, B, C, E



DFS Traversal

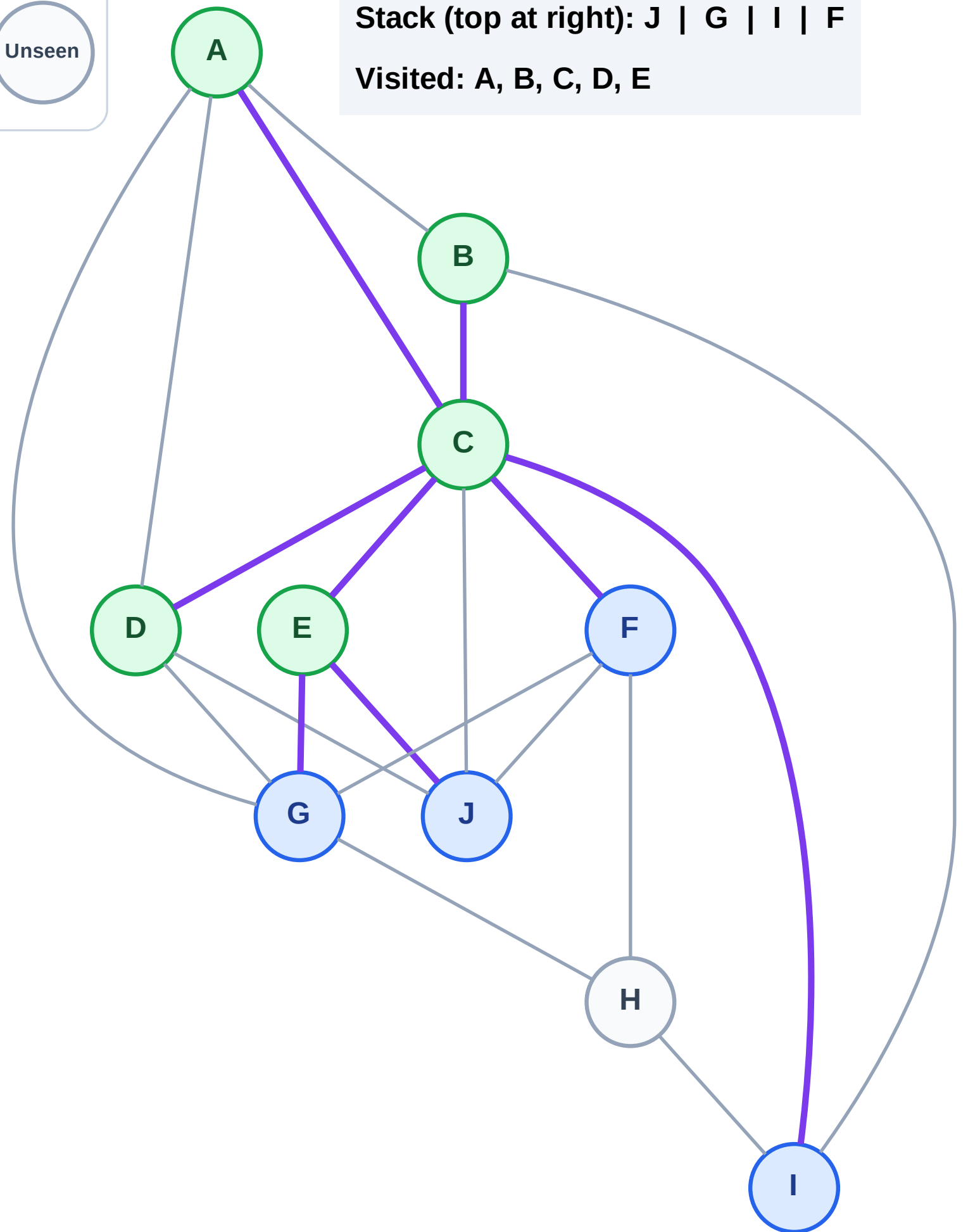
Step 11: No new neighbors from D

Legend



Stack (top at right): J | G | I | F

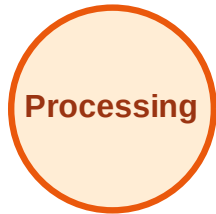
Visited: A, B, C, D, E



DFS Traversal

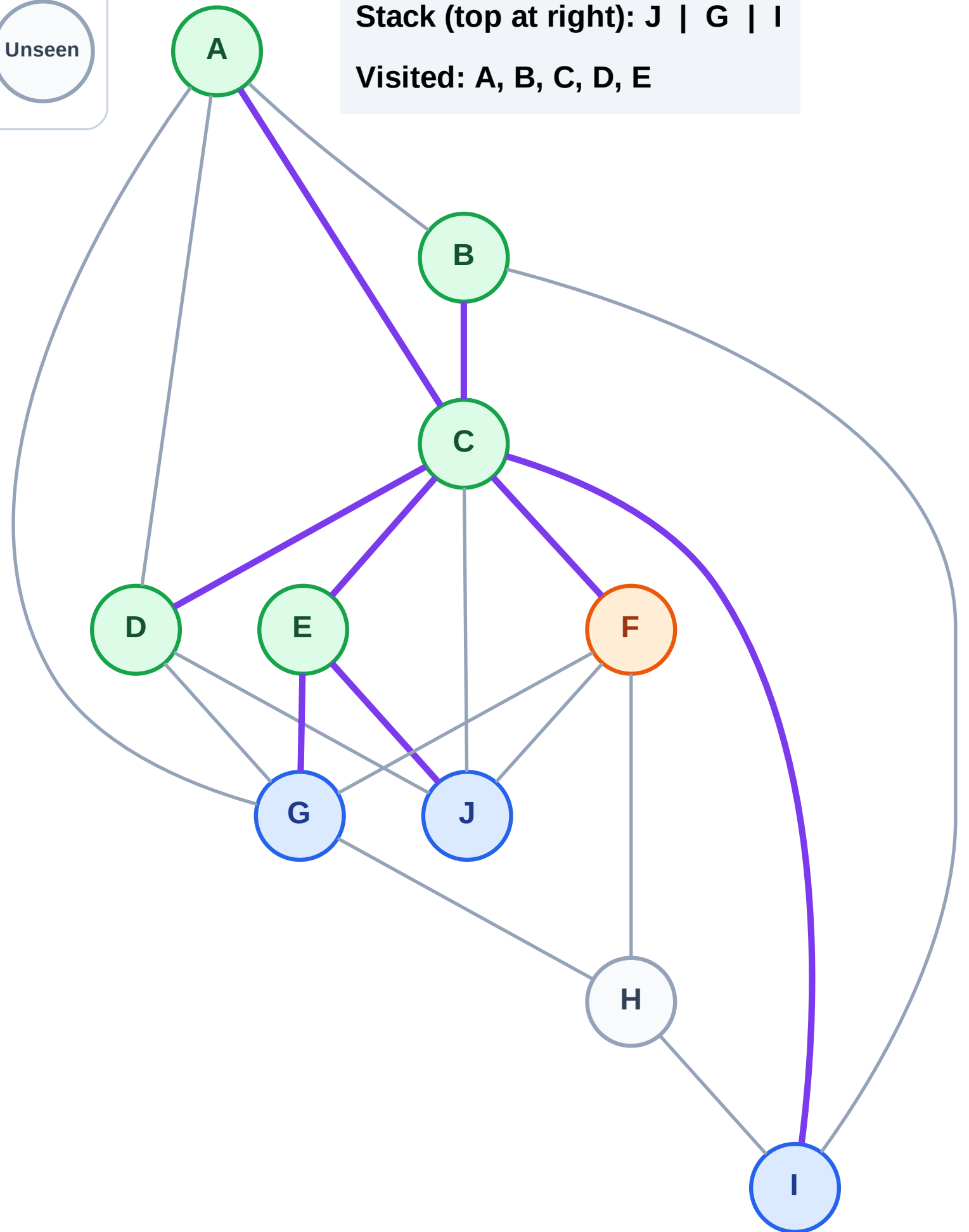
Step 12: Process node F

Legend



Stack (top at right): J | G | I

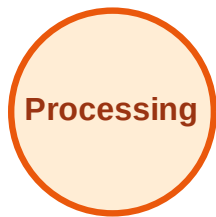
Visited: A, B, C, D, E



DFS Traversal

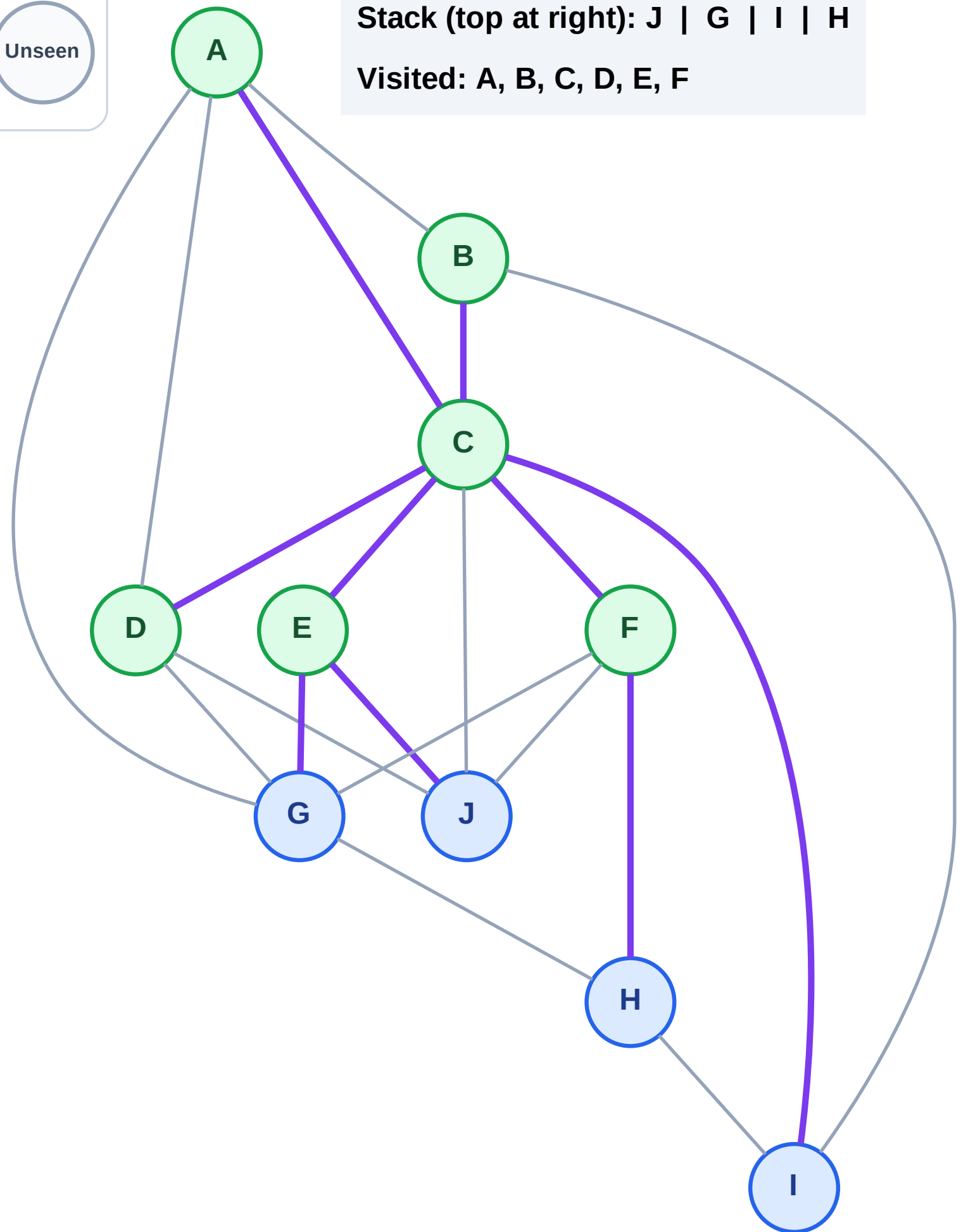
Step 13: Discover H from F

Legend



Stack (top at right): J | G | I | H

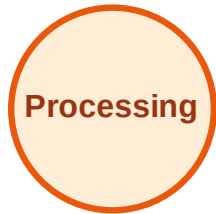
Visited: A, B, C, D, E, F



DFS Traversal

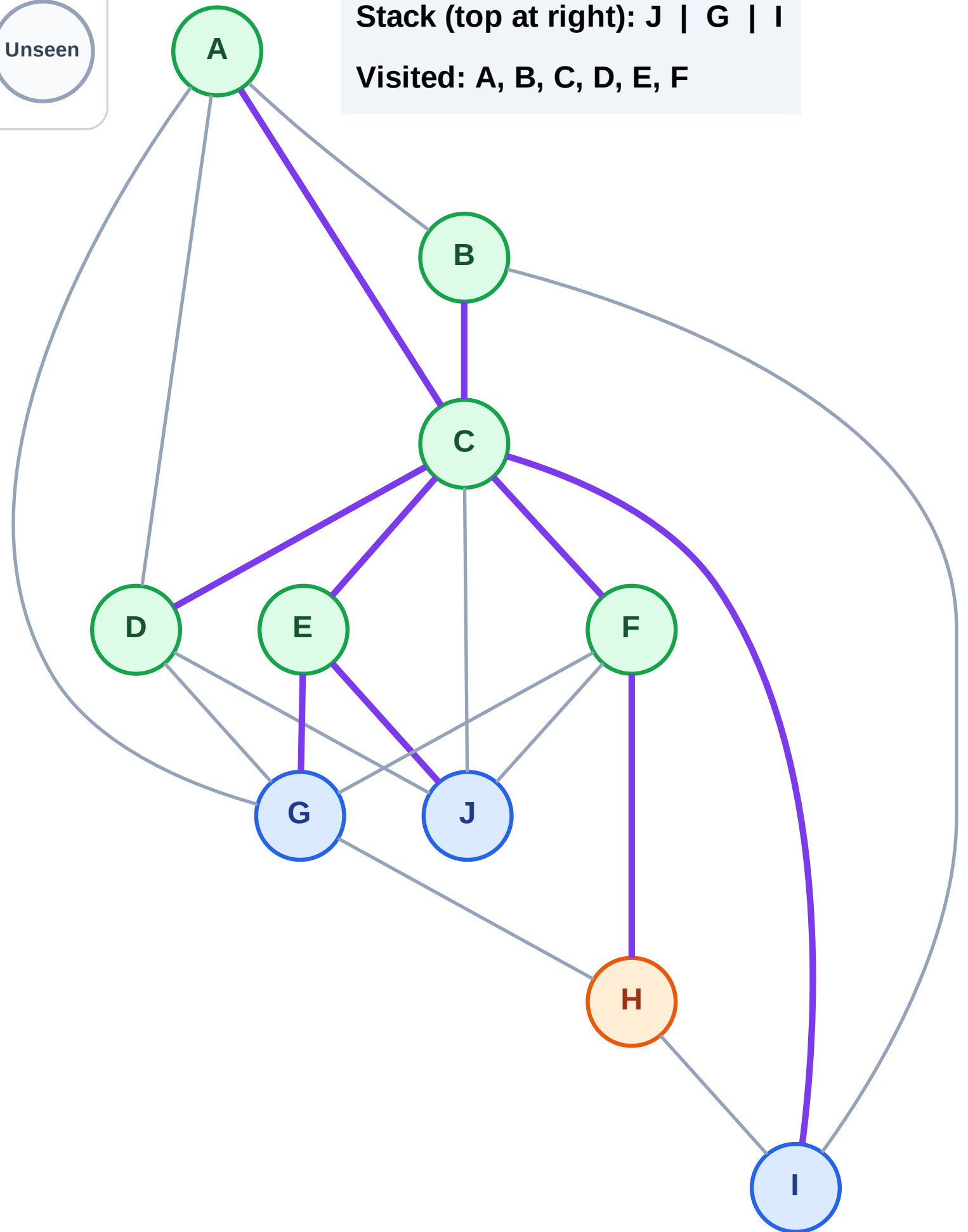
Step 14: Process node H

Legend



Stack (top at right): J | G | I

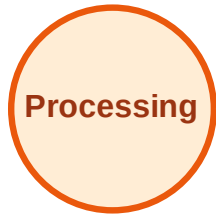
Visited: A, B, C, D, E, F



DFS Traversal

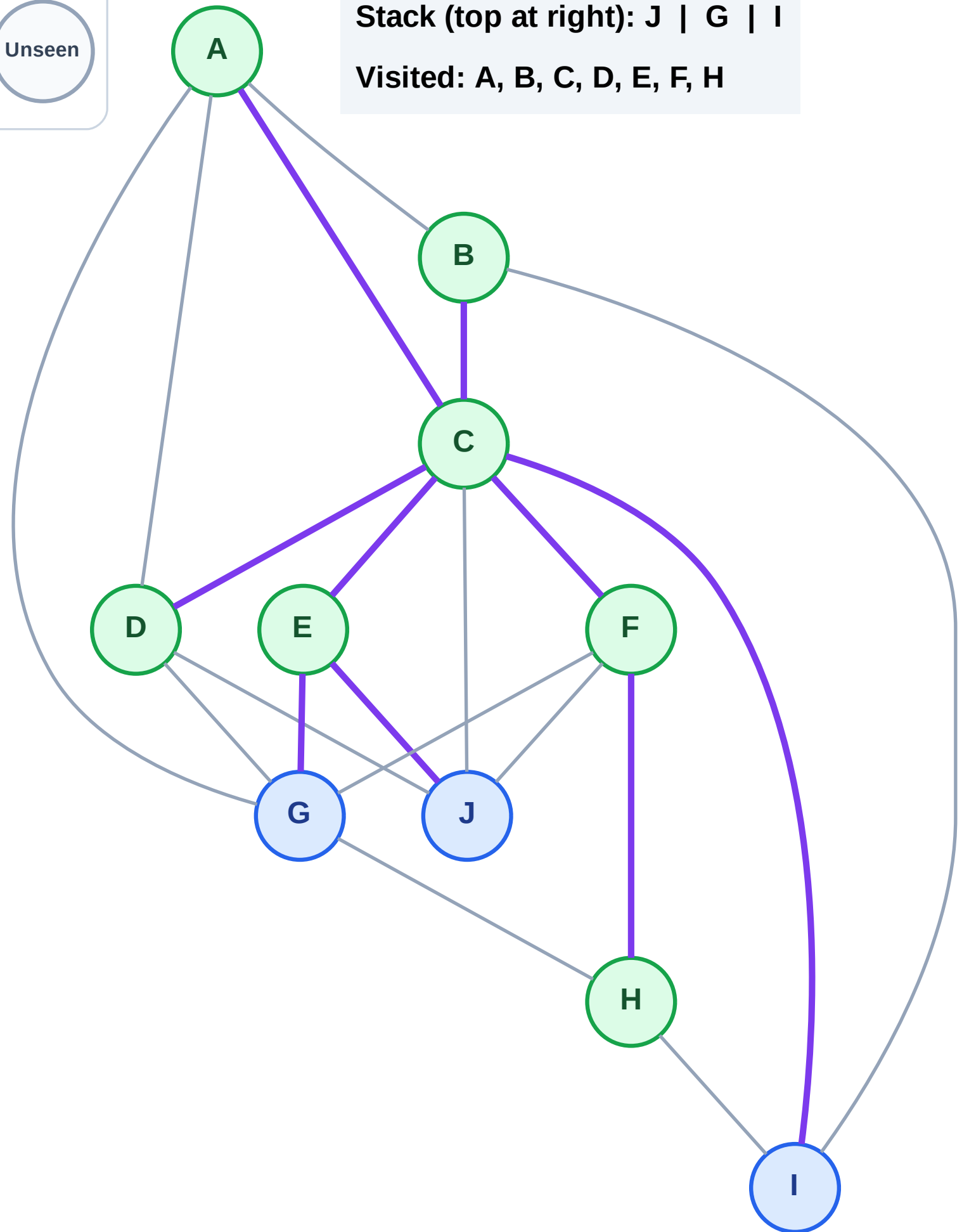
Step 15: No new neighbors from H

Legend



Stack (top at right): J | G | I

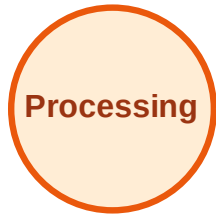
Visited: A, B, C, D, E, F, H



DFS Traversal

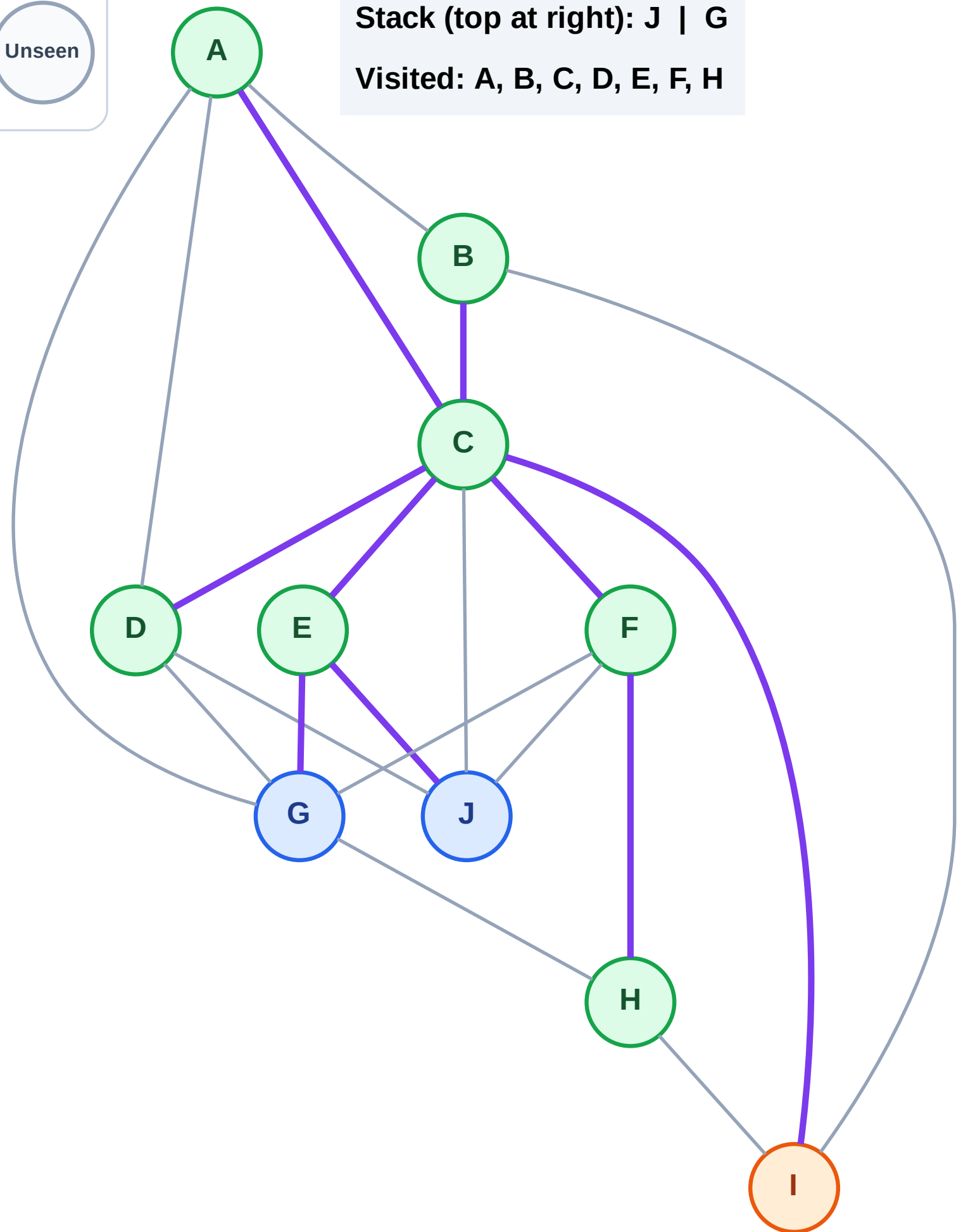
Step 16: Process node I

Legend



Stack (top at right): J | G

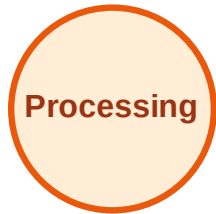
Visited: A, B, C, D, E, F, H



DFS Traversal

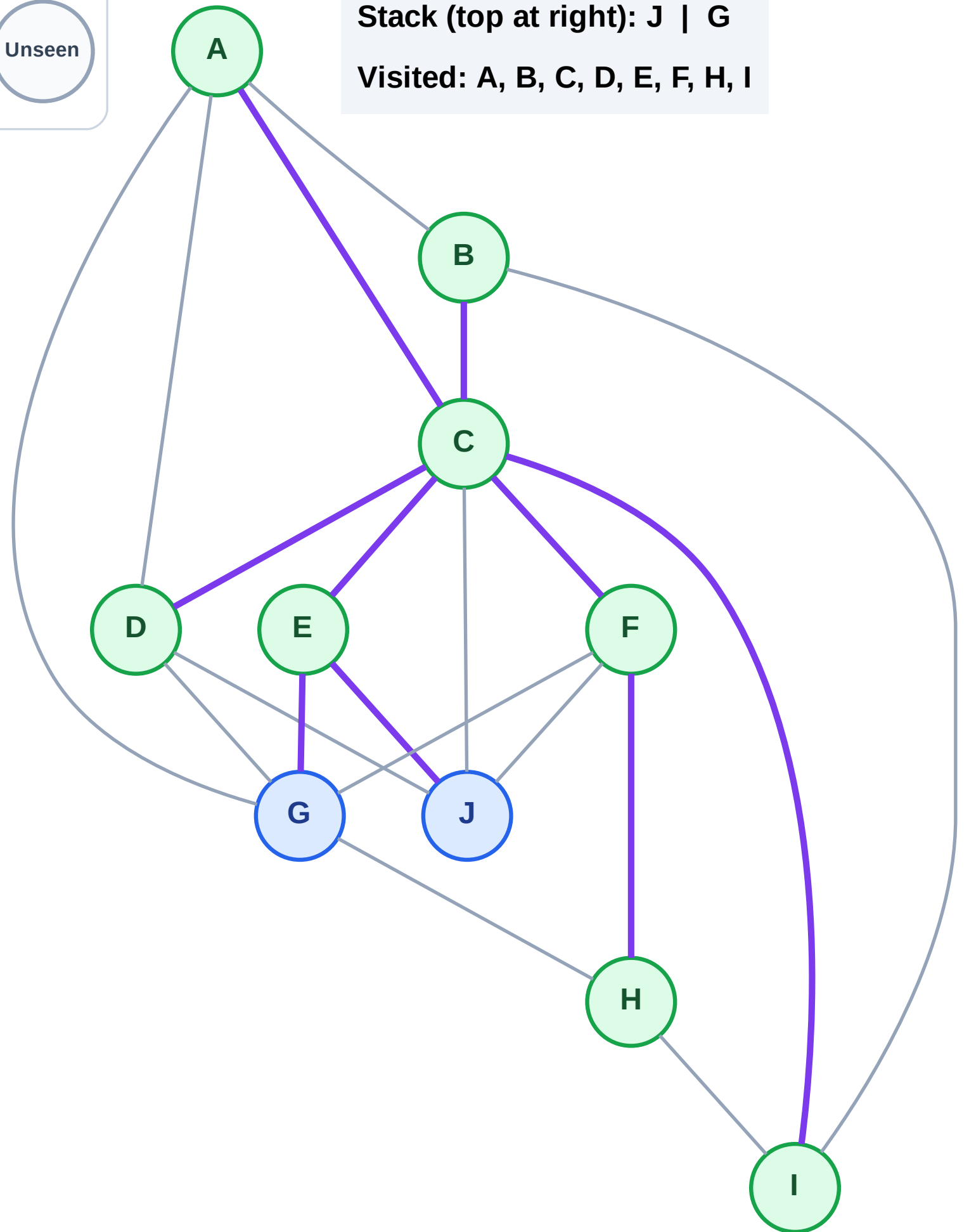
Step 17: No new neighbors from I

Legend



Stack (top at right): J | G

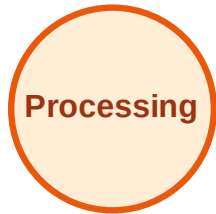
Visited: A, B, C, D, E, F, H, I



DFS Traversal

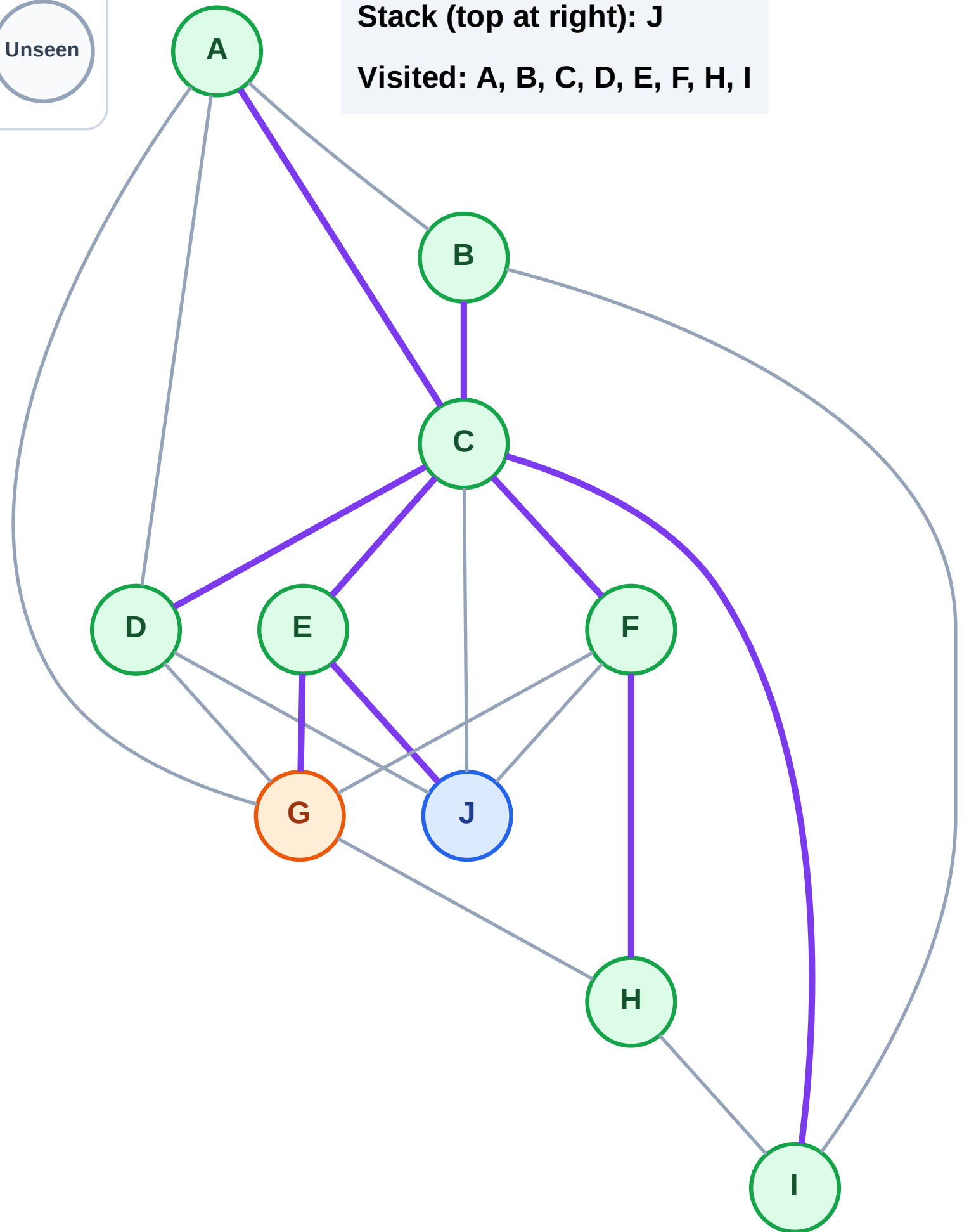
Step 18: Process node G

Legend



Stack (top at right): J

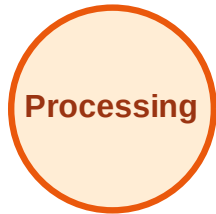
Visited: A, B, C, D, E, F, H, I



DFS Traversal

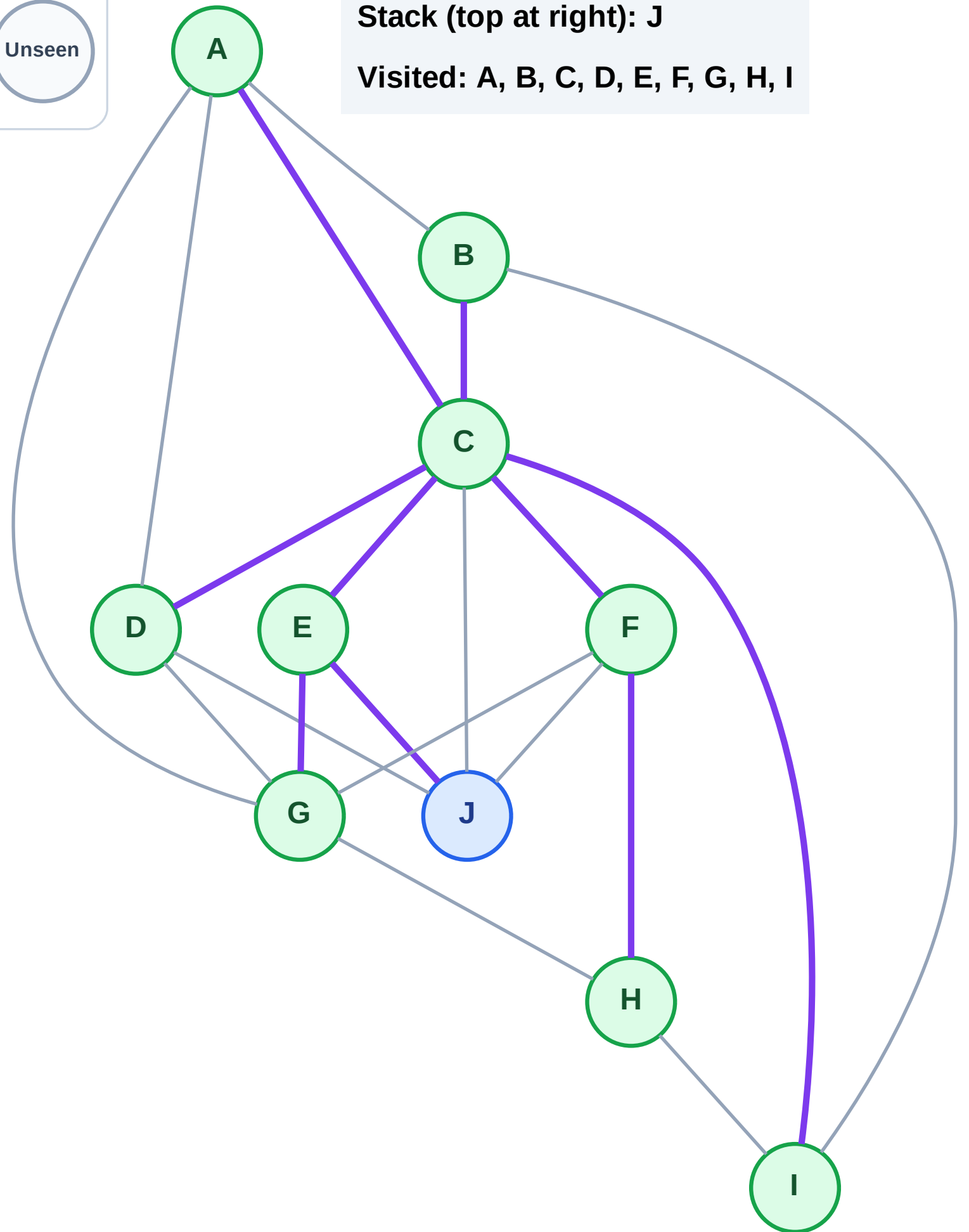
Step 19: No new neighbors from G

Legend



Stack (top at right): J

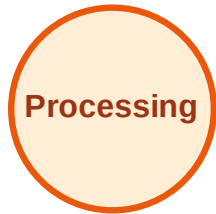
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

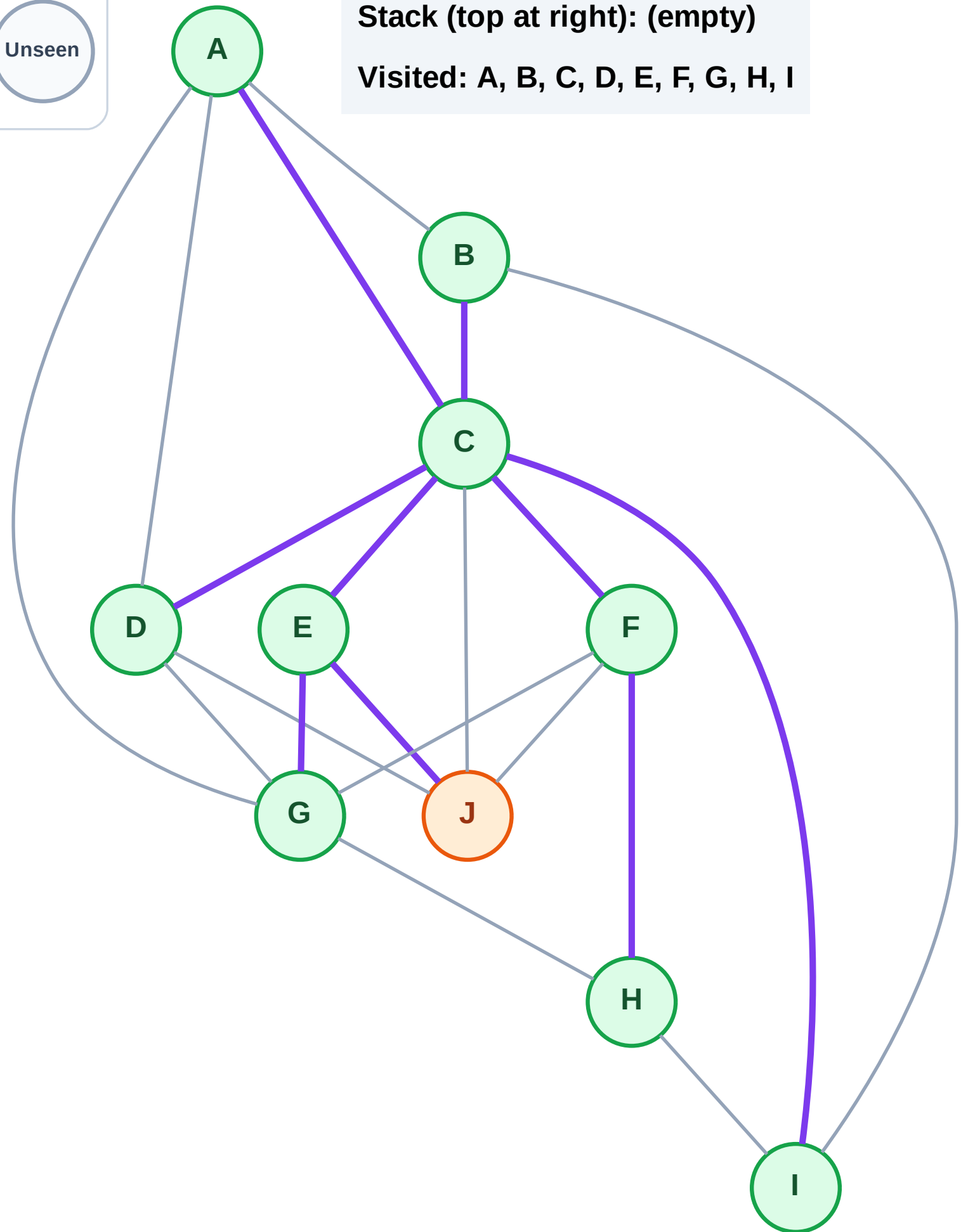
Step 20: Process node J

Legend



Stack (top at right): (empty)

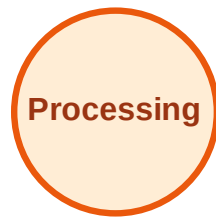
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

